

NL-PAC: Specification Ambiguity and Certified Minimax Risk Floors in LLM-Mediated Supervision

Berkay Anahtarci

Department of Mathematical Engineering, Özyeğin University, Istanbul, Türkiye
berkay.anahtarci@ozyegin.edu.tr

Abstract

Large language models increasingly provide labels, evaluations, and feedback for tasks specified in natural language. When a specification admits multiple readings but the supervision channel does not reveal which is operative, additional labels reduce sampling error without resolving the resulting identification problem. We introduce Natural Language PAC (NL-PAC), a framework that uses a fixed model’s thresholded decoding law to define admissible labels and candidate targets. The probability that multiple labels are admissible equals the diameter of the pointwise-admissible target class, and under target-blind supervision every learner incurs worst-case risk of at least half this diameter, at every sample size; the exact randomized minimax risk over this class is attained by a data-independent strategy. Finite-sample confidence bounds make these quantities certifiable from held-out unlabeled inputs. In a frozen Qwen 2.5–3B audit, one prespecified prompt yields a positive model-relative certificate, whereas a paraphrase and exact-rule controls yield zero. A held-out bridge audit finds that supplied candidate reading clauses fail the admissibility condition needed to transfer the certificate to coherent readings. The guarantee is specific to the audited model, prompt, threshold, and input distribution; extending it to human interpretations requires external validation.

1 Introduction

Modern learning pipelines increasingly use large-language-model (LLM) judgments in place of human evaluation or labeling, including in LLM-as-a-judge systems (Liu et al., 2023b; Kim et al., 2024; Gu et al., 2026). Prompts and scoring rubrics specify these tasks in natural language and may admit multiple interpretations. The supervising model can adopt one interpretation without revealing which one governs its judgments. We call the supervision channel *target-blind* when its observation law does not reveal which reading is operative. This regime is common in practice: judge models are typically served through proprietary APIs whose internals are opaque to the analyst (La Malfa et al., 2024), so the operative reading is unobservable by construction. Target blindness refers to this hidden reading; whether the model’s decoding probabilities are exposed is a separate matter that later governs the audit’s access mode. The resulting error is interpretive rather than statistical: additional observations from the same unresolved channel do not identify the operative reading, although a more informative channel can. We ask how much minimax risk remains when the channel defining the admissible targets is also the learner’s only source of supervision, and whether that risk floor can be certified from unlabeled inputs. We formalize the setting as *Natural Language PAC* (NL-PAC), characterize when such a channel creates an irreducible risk floor, and turn that floor into a finite-sample certificate.

Classical statistical learning theory does not model this coupling between task specification and the supervision channel: in the Probably Approximately Correct (PAC) framework of Valiant (1984),

the analyst fixes the instance space, distribution, target, and hypothesis class, and the learner pays only the statistical price of estimating a target inside that specification. This is the separation underlying VC theory and computational learning theory generally (Kearns and Vazirani, 1994). Language-model systems collapse this separation: the task is given as a natural-language instruction, and the supervision signal is produced by a model interpreting that same instruction (Liu et al., 2023a). NL-PAC makes this specification–channel pair the object of analysis.

Formally, a fixed model, prompt, and threshold induce at each input an *admissible label set* $A_\tau(x)$; the overlap mass $D_\tau^* = \mathbb{P}(|A_\tau(X)| \geq 2)$ is the diameter of the pointwise-admissible core of the induced candidate class, and under target-blind supervision every learner incurs worst-case risk at least $D_\tau^*/2$, with the exact obstruction given by a multiplicity-weighted value V_τ^* over the admissible core. This bound does not vanish with sample size. Because this floor is a functional of the model-induced admissible sets, it can be certified from unlabeled deployment inputs whenever those sets are observed exactly or estimated with a certified decoding radius. The rest of the paper develops and audits these quantities.

Contributions. We establish four results that together characterize how a model-induced ambiguity set creates an irreducible learning floor and how that floor can be audited from held-out unlabeled inputs.

1. **Representation.** Thresholded model outputs define a pointwise-admissible core and its open ζ -tolerance class. The core diameter equals the overlap mass $D_\tau^* = \mathbb{P}(|A_\tau(X)| \geq 2)$, while the tolerance-class diameter lies in $[D_\tau^*, D_\tau^* + 2\zeta]$, with both endpoints sharp (Theorem 2.11; sharpness in Section B).
2. **Blind-channel value.** Target blindness yields a half-diameter minimax floor over the tolerance class and the exact value V_τ^* over the core, uniformly in the learner’s sample size (Theorems 3.2 and 3.5). The latter is attained by a data-independent randomized learner and witnessed by an explicit finite least-favorable family. It equals $D_\tau^*/2$ when at most two labels are admissible at almost every input and is strictly sharper when higher-order overlap has positive mass.
3. **Coherent readings.** An explicit η -coverage condition bounds the gap between V_τ^* and the blind minimax value over a finite family of global readings (Theorem 3.9). This condition addresses the fact that pointwise selectors need not represent coherent global interpretations. A fit–holdout audit finds that the bound is uninformative for the supplied two-reading pool (Section 5).
4. **Certification.** Observed admissible sets give Hoeffding certificates for the pairwise floor, diameter, and exact core value: N held-out inputs give a pairwise lower certificate at a Hoeffding radius, and under sampled decoding the radius additionally pays threshold-margin and finite-depth terms (Theorem 4.1; Theorem 4.5; Theorem 4.6; Theorems 4.9 and 4.10).

Adjacent literatures. Two adjacent literatures bracket the problem without containing it. Noisy and weak supervision assume an operative target and analyze corrupted, aggregated, or heuristic observations of it (Natarajan et al., 2013; Ratner et al., 2017; Dawid and Skene, 1979). In classical malicious- and nasty-noise models, the resulting lower bounds scale with the corruption budget and vanish as that budget vanishes (Kearns and Li, 1993; Bshouty et al., 2002). Instance-dependent noise further illustrates the role of structural restrictions: bounded instance- and label-dependent noise is learnable under additional conditions, as shown by Cheng et al. (2020), whereas an unrestricted noise law leaves the target–noise decomposition unidentified. Truth-tracking analyses similarly ask

when reports from imperfect sources identify an externally defined state (Singleton and Booth, 2024).

LLM-as-a-judge research instead makes language part of the evaluation interface: work on rating indeterminacy documents that the criteria supplied to a judge may themselves admit multiple valid readings (Guerdan et al., 2025). Quantitative analyses bound how far judge labels can substitute for ground truth under *exogenous* judge error or bias (Dorner et al., 2025; Feuer et al., 2026). NL-PAC isolates the intersection of these concerns: the same specification-mediated channel both defines the admissible targets and supplies the observations, so its inability to reveal the operative reading creates an identification obstruction. That obstruction is set by the indistinguishability of admissible readings, not by a noise budget, and is endogenous to the model’s own interpretation.

On the alignment side, an information-theoretic argument holds that no specification can pin down an external target while conditional entropy remains (Young, 2025), and reward-misspecification studies show that optimizing a proxy for an underspecified objective can diverge sharply from the intended one (Pan et al., 2022); the ambiguity diameter certified here is an operational, model-relative witness of that gap. The closest prior quantity is the *ambiguity degree* of partial-label learning (Cour et al., 2011; Liu and Dietterich, 2014), which marks exactly the condition under which candidate-label ambiguity does *not* create an irreducible floor; NL-PAC characterizes the complementary regime, in which the channel cannot disambiguate. Further bordering literatures, including learning from disagreement, perspectivist and prompt-underspecification research, and inference on partially identified parameters, are treated in Section D.

Relation to classical theory. NL-PAC instantiates classical statistical decision theory in a language-mediated supervision problem. The two-target and least-favorable-prior reductions follow standard arguments (Wald, 1950; Blackwell and Girshick, 1954; Ferguson, 1967), and the per-input value coincides with the finite zero–one subset game from adversarial multiclass classification (Fathony et al., 2016). The NL-PAC contribution is to derive the target set and observation channel from a fixed model–prompt configuration, construct an explicit finite least-favorable family of admissible selectors, and make the resulting minimax value certifiable from held-out unlabeled inputs. The proofs combine classical two-point testing and concentration arguments (Tsybakov, 2009; Hoeffding, 1963) with this model-induced structure.

Empirical validation. The principal experiment audits a frozen Qwen judge’s declared-label conditional first-token probabilities, where the certificate applies with its correction-free radius: one prespecified prompt yields a positive model-relative certificate, while a second paraphrase and the exact-rule controls yield zero (Section 5). Two audits delimit the claim: the bridge passes its fitted mixture-coverage check but fails its held-out admissibility check for the supplied two-reading pool, and the sampled-decoding mode is inconclusive at feasible depth because its finite-depth and threshold-margin corrections are vacuous. Controlled tasks recovering the prescribed zero and positive certificate cases, and a ChaosNLI calculation illustrating the gain from multiclass multiplicity, appear in Section C.

Scope of the guarantee. The certificate quantifies ambiguity relative to the audited tuple $(\text{LLM}_\theta, \Pi, \tau, P)$; it does not by itself identify the designer’s intended target or establish human task ambiguity. Extending the guarantee to coherent human readings or deployment settings therefore requires coverage and external construct-validity evidence.

Roadmap. Section 2 defines the admissible geometry; Section 3 derives the blind-channel values and coherent-reading bridge; Section 4 constructs finite-sample certificates; and Sections 5 and 6 report the empirical probes, limitations, and extensions.

2 NL-PAC Setup and Model-Admissible Geometry

This section distinguishes four sources of uncertainty: ambiguity in the task description, stochasticity in model decoding, error relative to a target, and indistinguishability induced by the supervision channel. The resulting quantities describe the audited model and channel rather than ambiguity in human interpretation.

2.1 Supervision and admissibility channels

We adopt the standard PAC setting. Let $(\mathcal{X}, \mathcal{F}_{\mathcal{X}})$ be a measurable instance space, let $\mathcal{Y}$ be a finite label space with the discrete σ -algebra, and let P be a probability measure on $\mathcal{X}$. The target is a measurable labeling function $f: \mathcal{X} \rightarrow \mathcal{Y}$. In the classical setting, supervision consists of direct observations of $f(x)$.

Here, by contrast, the target is specified through natural language. Let $\mathcal{T}$ denote the model’s token alphabet and equip the set $\mathcal{T}^*$ of finite token strings with the discrete σ -algebra. The spaces of task descriptions and justifications are measurable subsets $S_{\text{NL}}, J_{\text{NL}} \subseteq \mathcal{T}^*$, endowed with the corresponding subspace σ -algebras $\mathcal{F}_{S_{\text{NL}}}$ and $\mathcal{F}_{J_{\text{NL}}}$. All sets of strings used below are assumed measurable. A fixed task description $f_{\text{NL}} \in S_{\text{NL}}$, supplied independently of the learner, is intended to specify the target f .

Throughout, the admissibility channel interprets the description with a pretrained language model LLM_{θ} whose parameters θ are frozen. Since LLM_{θ} is generative, $\text{LLM}_{\theta}(\cdot | u)$ is a distribution over token-string *continuations* of a string $u \in \mathcal{T}^*$, not over the label set $\mathcal{Y}$. Two fixed maps bridge the gap: a *label-extraction kernel* (verbalizer)

$$V: \mathcal{T}^* \rightarrow \Delta(\mathcal{Y})$$

taking a generated string to a distribution over labels, where $\Delta(\mathcal{Y})$ is the probability simplex on $\mathcal{Y}$ (e.g. constrained decoding onto a verbalizer set $\{v_y\}_{y \in \mathcal{Y}}$, or parsing an answer field), and a *prompt template*

$$\Pi: S_{\text{NL}} \times \mathcal{X} \rightarrow \mathcal{T}^*,$$

taking a description and an instance to the prompt $\Pi(f_{\text{NL}}, x)$ presented to LLM_{θ} . The kernel induces the label probability

$$p_{\text{LLM}_{\theta}}(y | u) := \sum_{s \in \mathcal{T}^*} V(y | s) \text{LLM}_{\theta}(s | u).$$

For the fixed description, define the prompt-conditioned *admissibility score*

$$\pi_{\text{LLM}_{\theta}}(y | f_{\text{NL}}, x) := p_{\text{LLM}_{\theta}}(y | \Pi(f_{\text{NL}}, x)).$$

This score is a decoding probability, not a calibrated posterior over human interpretations. All label probabilities below refer to this quantity. The description f_{NL} and distribution P are fixed throughout; we suppress them from A_{τ} , D_{τ}^* , and Sel_{τ} , displaying them only where a result varies them (as in $\mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ and $\text{diam}_{\tau, \zeta}(f_{\text{NL}}, P)$).

Assumption (A1) (Measurability). *For every $y \in \mathcal{Y}$, the map $x \mapsto \pi_{\text{LLM}_\theta}(y \mid f_{\text{NL}}, x)$ is $\mathcal{F}_{\mathcal{X}}$ -measurable.*

Abstractly, the learner’s supervision is an *observed-label kernel* $x \mapsto Q(\cdot \mid x) \in \Delta(\mathcal{Y})$, the conditional law of the label it receives at input x . For the fixed-description base model studied here we restrict attention to a kernel Q that carries *no* operative-target argument; Section 3 later embeds this base model in a target-indexed family Q_f in which target dependence, and hence partial distinguishability, is possible. Target-independence is thus a modeling restriction of the base channel, not a property discovered later. The headline blind-channel results depend only on Q , not on how it is realized. We realize Q concretely through a justification oracle g and a decoder ρ , introduced next, so that $Q(\cdot \mid x) = \text{Law}(\rho(J, x) \mid f_{\text{NL}}, x)$ with $J \sim g(\cdot \mid f_{\text{NL}}, x)$; readers interested only in the floors may read (g, ρ) as one instantiation of Q .

Definition 2.1 (Justification oracle). The justification oracle g is a Markov kernel from $(S_{\text{NL}} \times \mathcal{X}, \mathcal{F}_{S_{\text{NL}}} \otimes \mathcal{F}_{\mathcal{X}})$ to $(J_{\text{NL}}, \mathcal{F}_{J_{\text{NL}}})$. On input (f_{NL}, x) it draws a justification $J \sim g(\cdot \mid f_{\text{NL}}, x)$, with realizations $j \in J_{\text{NL}}$.

Assumption (A2) (Conditional independence of oracle draws). *The inputs are i.i.d., $X_{1:m} \sim P^m$, and given $X_{1:m}$ the draws $J_i \sim g(\cdot \mid f_{\text{NL}}, X_i)$ are conditionally independent across i .*

This excludes dependence among repeated oracle outputs beyond that induced by the sampled inputs. We impose full conditional independence and do not analyze weaker dependence conditions such as martingale-difference or mixing assumptions.

Definition 2.2 (Justification decoder). The *justification decoder* is a deterministic map $\rho: J_{\text{NL}} \times \mathcal{X} \rightarrow \mathcal{Y}$ that reads a label off a justification–input pair. It is required to be measurable with respect to the product σ -algebra $\mathcal{F}_{J_{\text{NL}}} \otimes \mathcal{F}_{\mathcal{X}}$ on its domain and the discrete σ -algebra on $\mathcal{Y}$, so that the composed observed label $\rho(J, X)$ is a well-defined $\mathcal{Y}$ -valued random variable.

The tuple $(\mathcal{X}, \mathcal{Y}, P, f, f_{\text{NL}}, g, \rho)$ defines an *NL-PAC learning problem*. The target labels are unobserved; instead, for $i = 1, \dots, m$ the learner receives a sample generated by drawing $X_i \sim P$, then a justification $J_i \sim g(\cdot \mid f_{\text{NL}}, X_i)$, and setting $Y_i = \rho(J_i, X_i)$. A *learner* is any measurable, possibly randomized rule that maps $(X_i, Y_i)_{i=1}^m$ to a measurable classifier $h: \mathcal{X} \rightarrow \mathcal{Y}$. We impose no hypothesis-class restriction, and measure its performance against the latent target by the standard risk $\text{err}_P(h) = \mathbb{P}_{X \sim P}[h(X) \neq f(X)]$.

The framework therefore separates two channels. The *admissibility channel* $x \mapsto \pi_{\text{LLM}_\theta}(\cdot \mid f_{\text{NL}}, x)$ determines which labels the model regards as compatible with the description, whereas the *supervision channel* (g, ρ) determines the labels available to the learner. The oracle g may be instantiated by the same model or by a distinct supervisor, so the two induced label distributions need not coincide. NL-PAC asks whether the observed channel supports low target risk or leaves an irreducible gap that cannot be closed by changing the learner or increasing the sample size.

Example 2.3 (Content moderation). Let $\mathcal{X}$ be user comments and $\mathcal{Y} = \{0, 1\}$ denote *safe* and *toxic*, with $f(x) = 1$ exactly when x contains a literal lexical insult. The description $f_{\text{NL}} = \text{“label the comment toxic (1) or safe (0)”}$ is silent on sarcasm, admitting two readings: (i) pragmatic personal attacks, including sarcasm, are toxic and (ii) only literal lexical insults are toxic, with the target following reading (ii). On $x = \text{“That was a brilliant move, genius”}$ we have $f(x) = 0$, yet the oracle may return $j = \text{“a sarcastic personal attack”}$, yielding $\rho(j, x) = 1 \neq f(x)$: not a decoding failure but the label under reading (i).

For a fixed description f_{NL} , the *supervision channel* (g, ρ) is conditioned on the input x , but not on the operative target. The resulting observation law therefore cannot distinguish among candidate targets associated with the same description.

Proposition 2.4 (Fixed-description target blindness). *Fix P , the description f_{NL} , the oracle g , and the decoder ρ , and draw $X \sim P$ and $J \sim g(\cdot \mid f_{\text{NL}}, X)$. The law of the observed pair $(X, \rho(J, X))$ does not depend on the operative target: it is identical under any two candidates $f_1, f_2: \mathcal{X} \rightarrow \mathcal{Y}$.*

Proof. The law of $(X, \rho(J, X))$ factors as the input marginal P composed with the observed-label kernel $Q(\cdot \mid x) = \text{Law}(\rho(J, x) \mid f_{\text{NL}}, x)$ with $J \sim g(\cdot \mid f_{\text{NL}}, x)$. This kernel is determined by $g(\cdot \mid f_{\text{NL}}, x)$ and the deterministic decoder $\rho(\cdot, x)$ alone; neither depends on the operative target, so Q , and hence the joint law, is identical under any two candidates f_1, f_2 . This is the sense in which the blind-channel results depend only on Q . $\square$

We call an observation channel with this property *target-blind*, and all subsequent blind-channel guarantees refer to this fixed-description observation law. *Oracle accuracy*, measured by the error rate

$$\epsilon_g := \mathbb{P}_{X \sim P, J \sim g(\cdot \mid f_{\text{NL}}, X)}[\rho(J, X) \neq f(X)],$$

is a separate property of the supervision channel, required neither for target blindness nor for the admissibility geometry; its compatibility with a localized unresolved floor is discussed in Theorem B.2. Thus the channel can be accurate on average yet uninformative exactly where candidate targets disagree.

This pins down the four sources of uncertainty announced above: description ambiguity surfaces as the admissible-overlap mass D_τ^* (Theorem 2.6); decoding stochasticity is the sampling controlled later by the plug-in analysis (Section 4.3); error relative to a target is ϵ_g above; and channel-induced indistinguishability is target blindness (Theorem 2.4), the property that turns D_τ^* into an irreducible floor.

2.2 Model-admissible labeling geometry

We quantify model-admissible ambiguity through the diameter of a class of labelings supported by the model's decoded probabilities. Two parameters enter the construction: the probability threshold τ determines which labels are admissible at an input, while the tolerance ζ controls the fraction of inputs on which a labeling may violate pointwise admissibility. Throughout, we take $\tau, \zeta \in (0, 1/2)$. The resulting diameter will reduce to an observable per-instance overlap statistic that drives both the risk floors and their empirical certificates.

Definition 2.5 (Admissible label set). For each $x \in \mathcal{X}$, the *admissible label set* at threshold τ is

$$A_\tau(x) := \{y \in \mathcal{Y} : \pi_{\text{LLM}_\theta}(y \mid f_{\text{NL}}, x) \geq \tau\}.$$

Thus a label is admissible precisely when its decoded probability is at least τ .

Assumption (A3) (Finite labels and nonempty admissible sets). *The label space satisfies $2 \leq |\mathcal{Y}| < \infty$, and $A_\tau(x)$ is nonempty for P -almost every x .*

This assumption ensures that pointwise admissible selections exist. Because every distribution on $\mathcal{Y}$ assigns probability at least $1/|\mathcal{Y}|$ to some label, nonemptiness is automatic when $\tau \leq 1/|\mathcal{Y}|$. In particular, it is automatic for binary labels under the standing condition $\tau < 1/2$.

Definition 2.6 (Admissible-overlap mass). The *admissible-overlap mass* is

$$D_\tau^* := \mathbb{P}_{x \sim P}[|A_\tau(x)| \geq 2].$$

Remark 2.7 (Role of the threshold τ). The threshold τ controls the strength of evidence required for a label to be retained. Multiple labels can be admissible only when $\tau \leq 1/2$; choosing $\tau < 1/2$ therefore allows overlap without requiring an exact tie. The geometric identities remain valid at other thresholds whenever Assumption (A3) holds, but $D_\tau^* = 0$ for $\tau > 1/2$. Moreover, admissible sets contract monotonically with the threshold:

$$\tau_1 \leq \tau_2 \implies A_{\tau_2}(x) \subseteq A_{\tau_1}(x) \implies D_{\tau_2}^* \leq D_{\tau_1}^*.$$

Hence increasing τ can only decrease the certified ambiguity floor. Our experiments avoid post hoc threshold selection by reporting a prespecified threshold sweep together with the corresponding certificate curve (Sections C.2 and 5).

Definition 2.8 (Model-admissible labeling class and diameter). (i) *Disagreement*. Call a measurable labeling $f': \mathcal{X} \rightarrow \mathcal{Y}$ *admissible at x* when $f'(x) \in A_\tau(x)$, and measure disagreement between two such labelings by the *disagreement pseudometric*

$$d_P(f_1, f_2) := \mathbb{P}_{x \sim P}[f_1(x) \neq f_2(x)].$$

(ii) *Class and core*. The *model-admissible labeling class* at tolerance ζ collects the labelings admissible off a set of mass at most ζ ,

$$\mathcal{F}_{\tau, \zeta}(f_{\text{NL}}) := \{f': \mathcal{X} \rightarrow \mathcal{Y} \text{ measurable} \mid \mathbb{P}_{x \sim P}[f'(x) \in A_\tau(x)] > 1 - \zeta\},$$

and its *almost-everywhere admissible core* is the zero-tolerance class, recovered as the intersection $\text{Sel}_\tau = \bigcap_{\zeta > 0} \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$,

$$\text{Sel}_\tau := \{f': \mathcal{X} \rightarrow \mathcal{Y} \text{ measurable} \mid f'(x) \in A_\tau(x) \text{ for } P\text{-a.e. } x\}.$$

(iii) *Diameter*. For a family $\mathcal{G}$ of labelings, write $\text{diam}_P(\mathcal{G}) := \sup_{f_1, f_2 \in \mathcal{G}} d_P(f_1, f_2)$, with $\text{diam}_P(\mathcal{G}) = 0$ when $|\mathcal{G}| \leq 1$. The *model-admissible diameter* is

$$\text{diam}_{\tau, \zeta}(f_{\text{NL}}, P) := \text{diam}_P(\mathcal{F}_{\tau, \zeta}(f_{\text{NL}})).$$

The disagreement pseudometric is standard in disagreement-based active learning (Hanneke, 2014). Since $\text{err}_P(h) = d_P(h, f)$, bounds expressed in d_P translate directly into PAC risk bounds.

These objects are well defined. For every $y \in \mathcal{Y}$,

$$\{x : y \in A_\tau(x)\} = \{x : \pi_{\text{LLM}_\theta}(y \mid f_{\text{NL}}, x) \geq \tau\}$$

is measurable by Assumption (A1). Since $\mathcal{Y}$ is finite, $x \mapsto A_\tau(x)$ is a measurable multifunction and every admissibility and disagreement event below is measurable; hence $\mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ and its diameter are well defined.

Conceptually, $\mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ is a model-relative identified set (Manski, 2003) formed from approximate measurable selections of A_τ (Molchanov and Molinari, 2018), and its core Sel_τ is a version space in the sense of Mitchell (1982), with model admissibility replacing labeled data as the defining constraint. What the analysis below exploits is that this identified set is model-relative and that its diameter reduces to a single auditable statistic.

For a measurable $u: \mathcal{X} \rightarrow \mathcal{Y}$, write $d_P(u, \text{Sel}_\tau) := \inf_{v \in \text{Sel}_\tau} d_P(u, v)$ for its distance to the admissible core Sel_τ .

Definition 2.9 (Maximal-spread pair). Fix a total order $\prec$ on $\mathcal{Y}$, and set $y_0 := \min_{\prec} \mathcal{Y}$. When $A_\tau(x) \neq \emptyset$, enumerate its elements $y_{(1)}(x), \dots, y_{(|A_\tau(x)|)}(x)$ in decreasing order of $\pi_{\text{LLM}_\theta}(\cdot \mid f_{\text{NL}}, x)$, with ties broken by $\prec$. The *maximal-spread pair* takes the two top-ranked admissible labels,

$$f_\tau^+(x) := y_{(1)}(x), \quad f_\tau^-(x) := y_{(\min\{2, |A_\tau(x)|\})}(x),$$

and both default to y_0 where $A_\tau(x) = \emptyset$. The two therefore coincide exactly when $|A_\tau(x)| \leq 1$.

In Theorem 2.3, the maximal-spread pair splits exactly on the sarcastic borderline comments: there both labels clear the threshold, $A_\tau(x) = \{0, 1\}$, and f_τ^+ takes the higher-scoring one, f_τ^- the other, while the two coincide on unambiguous comments; D_τ^* is the P -mass of the borderline region.

Lemma 2.10 (Maximal-spread identity). *Under Assumptions (A1) and (A3), the pair f_τ^+, f_τ^- lies in the admissible core Sel_τ and disagrees exactly on the overlap region,*

$$\{x : f_\tau^+(x) \neq f_\tau^-(x)\} = \{x : |A_\tau(x)| \geq 2\}.$$

In particular, $d_P(f_\tau^+, f_\tau^-) = D_\tau^$.*

Proof. By Assumption (A1) and finiteness of $\mathcal{Y}$ the ranked selectors are measurable, and both pick admissible labels wherever $A_\tau(x) \neq \emptyset$, which holds P -a.e. by Assumption (A3); hence $f_\tau^+, f_\tau^- \in \text{Sel}_\tau$. By construction they differ at x precisely when $A_\tau(x)$ holds at least two labels, so the displayed identity holds and its P -measure is D_τ^* . $\square$

The theorem below makes the diameter tractable: $\mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ is the open d_P -neighborhood of its core Sel_τ , and its diameter is bracketed by the overlap mass.

Theorem 2.11 (Geometry of the model-admissible labeling class). *Under Assumptions (A1) and (A3),*

$$\mathcal{F}_{\tau, \zeta}(f_{\text{NL}}) = \{u : \mathcal{X} \rightarrow \mathcal{Y} \text{ measurable} : d_P(u, \text{Sel}_\tau) < \zeta\}.$$

Moreover,

$$\text{diam}_P(\text{Sel}_\tau) = D_\tau^*, \quad D_\tau^* \leq \text{diam}_{\tau, \zeta}(f_{\text{NL}}, P) \leq D_\tau^* + 2\zeta.$$

Proof sketch; full proof in Section B. If u is within ζ of some $v \in \text{Sel}_\tau$, admissibility transfers from v to u off the small disagreement set, placing $u \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$. Conversely, an admissible $u \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ violates pointwise admissibility only on a set of mass below ζ ; reassigning u there to a top-ranked admissible label $y_{(1)}(x)$ yields a core member $v \in \text{Sel}_\tau$ that differs from u on that same set, so $d_P(u, v) < \zeta$. Hence $\mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ is exactly the open ζ -neighborhood of Sel_τ . Within the core, two selections can disagree only where $|A_\tau(x)| \geq 2$, and the maximal-spread pair attains that overlap exactly, giving $\text{diam}_P(\text{Sel}_\tau) = D_\tau^*$. For the upper bracket, apply this same reassignment to an arbitrary pair $u, v \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$, obtaining core members $\tilde{u}, \tilde{v} \in \text{Sel}_\tau$ with $d_P(u, \tilde{u}), d_P(v, \tilde{v}) < \zeta$; the triangle inequality then gives $d_P(u, v) \leq d_P(\tilde{u}, \tilde{v}) + d_P(u, \tilde{u}) + d_P(v, \tilde{v}) < D_\tau^* + 2\zeta$. $\square$

The additive 2ζ in the bracket is not slack of the analysis: both endpoints are attained, so no distribution-free bound improves it.

Proposition 2.12 (Tightness of the coverage bracket). *Both endpoints of the bracket $D_\tau^* \leq \text{diam}_{\tau, \zeta}(f_{\text{NL}}, P) \leq D_\tau^* + 2\zeta$ are sharp: each is realized as the diameter for some choice of P and admissible-set geometry:*

- (a) *if P restricted to $\{x : |A_\tau(x)| = 1\}$ is nonatomic and this set has mass at least 2ζ , then $\text{diam}_{\tau, \zeta}(f_{\text{NL}}, P) = D_\tau^* + 2\zeta$;*

(b) if P is purely atomic and every atom has probability at least ζ , then $\text{diam}_{\tau, \zeta}(f_{\text{NL}}, P) = D_\tau^*$.

Proof sketch; full proof in Section B. For (a), nonatomicity supplies disjoint subsets S_1, S_2 of the singleton region, each of mass $\zeta - \varepsilon$. Reassigning the maximal-spread pair to a distinct label on S_1 and S_2 keeps both labelings within coverage ζ yet adds $2(\zeta - \varepsilon)$ of disagreement beyond the overlap mass D_τ^* ; letting $\varepsilon \downarrow 0$ pushes the diameter to $D_\tau^* + 2\zeta$. For (b), an admissible labeling violates coverage on a set of mass below ζ , which can contain no atom once every atom has mass at least ζ ; that set is then P -null, so the ζ -neighborhood collapses onto the core and the diameter equals D_τ^* . $\square$

The identity $\text{diam}_P(\text{Sel}_\tau) = D_\tau^*$ equates the diameter of the almost-everywhere admissible core with the mass of inputs carrying multiple admissible labels. This ambiguity is a property of the model, prompt, and threshold, not of the learner’s sample: it persists under arbitrarily many further observations from the same target-blind channel.

3 Blind-Channel Risk Floors and Exact Minimax Values

This section characterizes the risk imposed by target-blind supervision. We first derive pairwise lower bounds that hold uniformly over the sample size, then identify the exact minimax value over the pointwise-admissible core and relate it to globally coherent readings. The admissible-overlap mass D_τ^* is the central geometric quantity.

3.1 Two-point blind-channel floor

We work in the *oracle-label model*, in which the learner observes labels decoded from oracle justifications rather than the target labels $f(X_i)$.

Definition 3.1 (Oracle-labeled sample). An *oracle-labeled sample* of size m is the sequence $S_m^{\text{oracle}} := ((X_i, \hat{Y}_i))_{i=1}^m$ obtained by drawing $X_{1:m} \sim P^m$, then drawing the justifications $J_1, \dots, J_m$ conditionally independently with $J_i \sim g(\cdot \mid f_{\text{NL}}, X_i)$, and setting $\hat{Y}_i := \rho(J_i, X_i)$.

The operative target enters the learner’s risk but not the sampling law above, so distinct targets can define the same statistical experiment. We turn this into a lower bound by Le Cam’s two-point method (Tsybakov, 2009): if two model-admissible labelings disagree on a set of mass D and the oracle cannot distinguish them there, the learner observes the same sample law under both targets, so no measurable rule can be correct for both.

Fix P , f_{NL} , g , and ρ , and consider two measurable candidate targets $f_1, f_2: \mathcal{X} \rightarrow \mathcal{Y}$. To accommodate both target-blind and target-dependent supervision, we momentarily allow the oracle to depend on the operative target: write $g(\cdot \mid f_{\text{NL}}, x, f)$ for a target-indexed family of Markov kernels (in the sense of Theorem 2.1 for each fixed f) that specializes to the target-blind oracle $g(\cdot \mid f_{\text{NL}}, x)$ of Theorem 2.1 exactly when it is constant in f . Let P_j denote the law of one observed example $(X, \rho(J_j, X))$ under target f_j , where $X \sim P$ and $J_j \sim g(\cdot \mid f_{\text{NL}}, X, f_j)$. Under Assumption (A2), a sample of size m has law $P_j^{\otimes m}$. Write $B := \{x : f_1(x) \neq f_2(x)\}$ for the disagreement region and $D := P(B) = d_P(f_1, f_2)$ for its probability. This pairwise construction does not require either target to be model-admissible.

Theorem 3.2 (Two-point blind-channel floor). *For every sample size m and every measurable, possibly randomized learner $\hat{h}$ mapping an oracle-labeled sample to a classifier, the following bounds hold.*

(a) (Le Cam bound.)

$$\max_{j \in \{1,2\}} \mathbb{E}_{S \sim P_j^{\otimes m}} [d_P(\hat{h}(S), f_j)] \geq \frac{D}{2} (1 - \text{TV}(P_1^{\otimes m}, P_2^{\otimes m})),$$

where the expectation averages over the sample and any internal randomization.

(b) (Unresolved-ambiguity floor.) Suppose the oracle does not distinguish the two targets: for P -almost every x ,

$$\text{Law}(\rho(J_1, x) \mid f_{\text{NL}}, x, f_1) = \text{Law}(\rho(J_2, x) \mid f_{\text{NL}}, x, f_2). \quad (1)$$

Then the two targets induce the same one-example observation law, $P_1 = P_2$, so $P_1^{\otimes m} = P_2^{\otimes m}$ and the bound of part (a) is independent of the sample size:

$$\max_{j \in \{1,2\}} \mathbb{E}_{S \sim P_j^{\otimes m}} [d_P(\hat{h}(S), f_j)] \geq \frac{D}{2} \quad \text{for every } m.$$

Proof sketch; full proof in Section B. For (a), any returned classifier satisfies $d_P(h, f_1) + d_P(h, f_2) \geq D$ pointwise on B , since $h(x)$ can match at most one of two distinct labels; integrating this against the common part $\min(p_1, p_2)$ of the two sample laws reduces the problem to testing $P_1^{\otimes m}$ against $P_2^{\otimes m}$ and yields the $\frac{D}{2}(1 - \text{TV})$ bound. For (b), condition Equation (1) equates the one-example observation laws, $P_1 = P_2$, hence $P_1^{\otimes m} = P_2^{\otimes m}$ and the total-variation term vanishes for every m . The bound of part (a) therefore reduces to the sample-independent floor $D/2$: no sample size recovers what the channel never transmits. $\square$

This blind-channel bound is exact for the two-target problem: a data-independent learner that returns f_1 or f_2 with equal probability has risk $D/2$ against either target, so the floor of part (b) is attained. The exactness is for the two-target minimax, not necessarily for the entire admissible class; Theorem B.1 records the general least-favorable-prior identity.

Corollary 3.3 (Diameter floor for the base channel). *Suppose that, for every target $f \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$, each observed example is generated by $X \sim P$, $J \sim g(\cdot \mid f_{\text{NL}}, X)$, and $\hat{Y} = \rho(J, X)$. Then, for every sample size m ,*

$$\inf_{\hat{h}} \sup_{f \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})} \mathbb{E}[d_P(\hat{h}(S_m^{\text{oracle}}), f)] \geq \frac{1}{2} \text{diam}_{\tau, \zeta}(f_{\text{NL}}, P).$$

The infimum ranges over all measurable, possibly randomized learners, and the expectation averages over the sample and the learner's internal randomization.

Proof. Every pair $f_1, f_2 \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ is admissible and, on the fixed-description base channel, unresolved (Theorem 2.4), so Theorem 3.2(b) gives $\max_{j \in \{1,2\}} \mathbb{E}[d_P(\hat{h}, f_j)] \geq \frac{1}{2} d_P(f_1, f_2)$ uniformly in m . Since $\sup_f \mathbb{E}[d_P(\hat{h}, f)]$ dominates this two-point maximum, optimizing the pair so that $d_P(f_1, f_2)$ approaches $\text{diam}_{\tau, \zeta}(f_{\text{NL}}, P)$ yields the bound; the supremum need not be attained (Theorem 2.12). $\square$

We call a pair *full-law unresolved* if its one-example observation laws coincide. Every pair has this property under the fixed-description channel, because the sampling law does not depend on the operative target. In the deterministic extreme, the target-blind channel always emits labels from one fixed reading f' , whose error against a target f is $d_P(f, f')$. The resulting $D/2$ floor is uniform over the sample size but remains a pairwise statement; it does not determine the minimax risk over the entire admissible class. We compute that full-class value next.

3.2 Exact blind minimax values

We now pass from a single unresolved pair to the full pointwise-admissible core Sel_τ . The resulting minimax value is determined by the number of admissible labels at each input. To define this multiplicity everywhere, fix $y_0 \in \mathcal{Y}$ and set

$$\bar{A}_\tau(x) := \begin{cases} A_\tau(x), & A_\tau(x) \neq \emptyset, \\ \{y_0\}, & A_\tau(x) = \emptyset, \end{cases} \quad k(x) := |\bar{A}_\tau(x)|.$$

By Assumption (A3), $\bar{A}_\tau(x) = A_\tau(x)$ for P -almost every x , so the completion is immaterial for all population quantities below.

Definition 3.4 (Admissible-multiplicity functional). The *expected non-modal admissible mass* is

$$V_\tau^\star := \mathbb{E}_{x \sim P} \left[1 - \frac{1}{k(x)} \right].$$

At each input x , uniform randomization over $\bar{A}_\tau(x)$ incurs error $1 - 1/k(x)$ against every admissible target label. Thus $V_\tau^\star$ is the population risk achieved by the uniform admissible strategy.

The three central quantities play different roles. The overlap mass $D_\tau^\star$ counts the inputs carrying more than one admissible label; the exact value $V_\tau^\star$ is the actual minimax error a blind learner must incur; and the diameter $\text{diam}_{\tau, \zeta}$ measures the spread of the admissible class. They coincide up to a factor of one half, $V_\tau^\star = \frac{1}{2} D_\tau^\star = \frac{1}{2} \text{diam}_P(\text{Sel}_\tau)$, exactly when at most two labels are admissible almost everywhere, and separate once higher-order overlap has positive mass.

We show next that the uniform admissible strategy is minimax optimal under target blindness.

Theorem 3.5 (Exact blind-channel minimax over admissible selections). *Under Assumptions (A1) and (A3), suppose the supervision channel is target-blind on Sel_τ (automatic for the fixed-description channel, Theorem 2.4): the one-example observation law P_f it induces under a target f is common to all $f \in \text{Sel}_\tau$. Then, for every sample size m ,*

$$\inf_{\hat{h}} \sup_{f \in \text{Sel}_\tau} \mathbb{E}_{S \sim P_f^{\otimes m}} [d_P(\hat{h}(S), f)] = V_\tau^\star,$$

the infimum ranging over all randomized measurable learners and the expectation including their internal randomness. The common value is attained by a data-independent learner.

Proof sketch; full proof in Section B. Achievability. Consider the data-independent learner given by the measurable Markov kernel $x \mapsto \text{Uniform}(\bar{A}_\tau(x))$ (a uniform draw from the finite cyclic selector family of the full proof, hence measurable since $\bar{A}_\tau$ is a measurable finite-valued multifunction). It errs against any fixed $f \in \text{Sel}_\tau$ with probability $1 - 1/k(x)$ at x ; its risk is therefore $V_\tau^\star$ against every target.

Lower bound. Target-blindness on Sel_τ makes the sample law identical across the class, so no learner's output depends on which target is operative. Averaging the risk over a finite cyclic family of admissible selections whose uniform mixture places uniform mass on $\bar{A}_\tau(x)$ at every x shows that no learner improves on $\mathbb{E}_x[1 - 1/k(x)] = V_\tau^\star$. The family is thus least favorable, and the two bounds meet for every sample size m . $\square$

For any target family $\mathcal{G}$, define its sample-size- m minimax risk by

$$R_m(\mathcal{G}) := \inf_{\hat{h}} \sup_{f \in \mathcal{G}} \mathbb{E}_{S \sim P_f^{\otimes m}} [d_P(\hat{h}(S), f)].$$

Object	Construction	Interpretation	Governs
Sel_τ (Def. 2.8)	pointwise admissible (P -a.e.)	exact theoretical core	exact minimax value V_τ^*
$\mathcal{F}_{\tau,\zeta}(f_{\text{NL}})$ (Thm. 2.11)	within ζ of the core	tolerance class	ζ -bracketed neighborhood minimax
$\mathcal{F}_{\tau,\zeta}^{\text{read}}(f_{\text{NL}})$ (Def. 3.6)	induced by a global reading clause	semantic candidate pool	coherent-reading bridge

Table 1: The three labeling classes and the result each governs.

The theorem above establishes $R_m(\text{Sel}_\tau) = V_\tau^*$ for every m , and its pointwise term $1 - 1/k(x)$ coincides with the subset value of the adversarial multiclass zero-one game of Fathony et al. (2016). The equality admits a classical decision-theoretic reading as well: the cyclic selector family is least favorable for an uninformative experiment in the sense of Blackwell and Girshick (1954). Together these two viewpoints furnish an explicit finite measurable selector family for the model-induced sets $\bar{A}_\tau(x)$, together with an aggregate value V_τ^* estimable from held-out unlabeled inputs.

The equality $R_m(\text{Sel}_\tau) = V_\tau^*$ is exact on the pointwise-admissible core Sel_τ , but transfers automatically neither to the wider ζ -neighborhood $\mathcal{F}_{\tau,\zeta}(f_{\text{NL}})$ nor to a class of globally coherent readings: the former tolerates admissibility violations on a set of small probability, while the latter constrains how labels are selected jointly across inputs. These are genuinely distinct decision problems and are treated separately.

3.3 From pointwise selectors to coherent readings

The defining condition of $\mathcal{F}_{\tau,\zeta}(f_{\text{NL}})$ constrains admissibility pointwise, so a labeling in $\mathcal{F}_{\tau,\zeta}(f_{\text{NL}})$ may vary its interpretation across inputs without corresponding to any single global reading. We therefore separate labelings induced by one reading clause from those assembled by pointwise selection. Table 1 summarizes the three labeling classes used in this paper.

Definition 3.6 (Coherent-reading subclass). Let $\mathcal{R}_{\text{read}}$ be a finite collection of natural-language *reading clauses*, each specifying an interpretation of f_{NL} . For $a \in \mathcal{R}_{\text{read}}$, let $\iota(f_{\text{NL}}, a)$ denote the description obtained by appending clause a to f_{NL} , and define its induced labeling by

$$f_a(x) := \arg \max_{y \in \mathcal{Y}} p_{\text{LLM}_\theta}(y \mid \Pi(\iota(f_{\text{NL}}, a), x)),$$

with ties resolved by a fixed ordering of $\mathcal{Y}$. We extend Assumption (A1) to every appended clause: for each $a \in \mathcal{R}_{\text{read}}$ and $y \in \mathcal{Y}$, the map $x \mapsto p_{\text{LLM}_\theta}(y \mid \Pi(\iota(f_{\text{NL}}, a), x))$ is $\mathcal{F}_{\mathcal{X}}$ -measurable. With $\mathcal{Y}$ finite and the tie-break fixed, each f_a is then a measurable labeling, so membership in $\mathcal{F}_{\tau,\zeta}(f_{\text{NL}})$ is well posed. The *coherent-reading subclass* consists of the induced labelings that satisfy the admissibility criterion:

$$\mathcal{F}_{\tau,\zeta}^{\text{read}}(f_{\text{NL}}) := \{f_a : a \in \mathcal{R}_{\text{read}}\} \cap \mathcal{F}_{\tau,\zeta}(f_{\text{NL}}).$$

The coherent-reading subclass is relative to the chosen clause collection and need not exhaust all coherent interpretations of f_{NL} . Enlarging $\mathcal{R}_{\text{read}}$ can only enlarge $\mathcal{F}_{\tau,\zeta}^{\text{read}}(f_{\text{NL}})$.

Remark 3.7 (What the diameter does and does not measure). *The inclusion $\mathcal{F}_{\tau,\zeta}^{\text{read}}(f_{\text{NL}}) \subseteq \mathcal{F}_{\tau,\zeta}(f_{\text{NL}})$ gives*

$$\text{diam}_P(\mathcal{F}_{\tau,\zeta}^{\text{read}}(f_{\text{NL}})) \leq \text{diam}_{\tau,\zeta}(f_{\text{NL}}, P) \leq D_\tau^* + 2\zeta.$$

The identity $D_\tau^ = \text{diam}_P(\text{Sel}_\tau)$ concerns the pointwise-admissible core. It does not imply that D_τ^* bounds the coherent-reading diameter, because members of the latter class may use their ζ coverage slack. Conversely, the maximal-spread pair attaining D_τ^* may switch labels across inputs and need*

not arise from any single global reading. Thus $D_\tau^\star$ measures pointwise model-admissible variation, not necessarily the spread among coherent interpretations.

We next compare the pointwise value $V_\tau^\star$ with the minimax value of a finite coherent family. If a channel is target-blind on a finite target class $\mathcal{T}$, then $R_m(\mathcal{T})$ is independent of m ; denote this common value by $V_{\text{blind}}(\mathcal{T}) := R_m(\mathcal{T})$. For a prior λ on a coherent family $\mathcal{C} = \{f_a\}$, let $\mu_x^\lambda(y) := \sum_{a: f_a(x)=y} \lambda_a$ denote the induced label distribution at input x .

Definition 3.8 (η -uniform coverage). Fix $\eta \geq 0$. A finite family $\mathcal{C} \subseteq \mathcal{F}_{\tau, \zeta}^{\text{read}}(f_{\text{NL}})$ is η -uniformly covering for Sel_τ if there exists a distribution $\lambda^\star \in \Delta(\mathcal{C})$ such that

$$\mathbb{E}_{x \sim P} \left[\max_{y \in \mathcal{Y}} \mu_x^{\lambda^\star}(y) - \frac{1}{k(x)} \right] \leq \eta.$$

Here η bounds the *average* gap between the mixture's modal concentration $\max_y \mu_x^{\lambda^\star}(y)$ and the uniform-admissible benchmark $1/k(x)$. This per-input gap can be negative where a reading places mass outside $A_\tau(x)$, so it is not a pointwise excess; only its average, together with ζ , controls the distance between the coherent and pointwise blind values. Smaller η indicates that the coherent family more closely reproduces the local spread of the pointwise-admissible core.

Theorem 3.9 (Coherent-reading bridge). *Let $\mathcal{C} \subseteq \mathcal{F}_{\tau, \zeta}^{\text{read}}(f_{\text{NL}})$ be finite and nonempty, and let the supervision channel be target-blind on $\mathcal{C}$. Then the coherent blind value never exceeds the core value by more than ζ ,*

$$V_{\text{blind}}(\mathcal{C}) \leq V_\tau^\star + \zeta,$$

and if $\mathcal{C}$ is moreover η -uniformly covering for Sel_τ , the two values agree up to $\max\{\eta, \zeta\}$:

$$V_\tau^\star - \eta \leq V_{\text{blind}}(\mathcal{C}) \leq V_\tau^\star + \zeta, \quad |V_\tau^\star - V_{\text{blind}}(\mathcal{C})| \leq \max\{\eta, \zeta\}.$$

Proof sketch; full proof in Section B. For the upper bound, consider the data-independent strategy that predicts uniformly on $\bar{A}_\tau(x)$. Against any $f_a \in \mathcal{C}$, its pointwise error is $1 - 1/k(x)$ wherever $f_a(x) \in A_\tau(x)$ and is at most one elsewhere. Since each f_a violates admissibility on a set of probability less than ζ , its risk is at most $V_\tau^\star + \zeta$.

For the lower bound, place the covering distribution $\lambda^\star$ on $\mathcal{C}$. Target blindness makes the learner's observation law independent of the sampled target. At input x , the smallest Bayes error against the induced label distribution $\mu_x^{\lambda^\star}$ is $1 - \max_y \mu_x^{\lambda^\star}(y)$. Therefore

$$V_{\text{blind}}(\mathcal{C}) \geq \mathbb{E}_x \left[1 - \max_y \mu_x^{\lambda^\star}(y) \right] = V_\tau^\star - \mathbb{E}_x \left[\max_y \mu_x^{\lambda^\star}(y) - \frac{1}{k(x)} \right] \geq V_\tau^\star - \eta.$$

□

Remark 3.10 (Reading the bridge). *The bound separates two approximation errors. The tolerance ζ measures departures from pointwise admissibility, whereas η measures how closely a mixture of coherent readings matches the balanced label distribution of the pointwise core. If both vanish, the coherent and pointwise blind values agree. The theorem does not ensure that a given reading pool has small coverage error. Both quantities are pool-relative and should be reported together: a small value of η is informative only when the corresponding admissibility tolerance ζ is also small.*

The lower bound is operational only if the coverage error of the supplied reading pool can be bounded from data. Sample splitting provides such a certificate. On a fitting sample $z_{1:N}^{\text{fit}}$, choose

$$\hat{\lambda} \in \arg \min_{\lambda \in \Delta(\mathcal{C})} \frac{1}{N} \sum_{i=1}^N \left(\max_y \mu_{z_i^{\text{fit}}}^\lambda(y) - \frac{1}{k(z_i^{\text{fit}})} \right).$$

Evaluate the same bounded loss at $\hat{\lambda}$ on an independent audit sample. Conditional on the fitting sample, $\hat{\lambda}$ is fixed, so a standard one-sided concentration bound yields a valid population upper bound on its coverage error. Reusing the fitting sample would instead require a uniform bound over λ .

4 Finite-Sample Certification of Blind-Channel Values

We construct finite-sample certificates from held-out inputs drawn from the deployment distribution P . Empirical admissible overlap certifies a pairwise risk floor and brackets the model-admissible diameter, while admissible-set multiplicities certify the exact core value V_τ^* . The latter can be strictly sharper when more than two labels overlap. These guarantees are specific to the audited deployment distribution.

The risk certificates are one-sided. A positive lower bound establishes a minimax floor; a small or zero bound is inconclusive, because sampling uncertainty may dominate the observed disagreement or the audited pair may not witness the full class diameter.

The section builds several certificates from a single overlap audit, and their scope varies along two axes: the *access mode* (exposed decoding probabilities versus sampled decoding) and the *target quantity* (the pairwise blind floor, the model-admissible diameter, or the exact core value V_τ^*). The reader may use the following map. Under exposed probabilities, Theorem 4.1 certifies the diameter band and the pairwise floor; Theorem 4.6 packages the pairwise floor for the canonical maximal-spread pair into the headline canonical-pair certificate; and Theorem 4.9 sharpens this to the exact value V_τ^* using admissible-set multiplicities. Theorem B.4 lifts the pairwise floor to a finite family by a union bound. Under sampled decoding, the same three quantities are certified with an enlarged plug-in radius (Theorem 4.5; Theorem 4.10). Algorithm 4.1 collects the exposed and sampled procedures into one audit.

4.1 Estimating model-admissible overlap

The admissible-overlap mass satisfies $D_\tau^* = \mathbb{E}[\mathbf{1}\{|A_\tau(X)| \geq 2\}]$. Its Bernoulli integrand records whether the model admits multiple labels at an input and is observable from the description and input alone. Consequently, D_τ^* can be estimated from unlabeled deployment inputs at the usual $N^{-1/2}$ rate.

Proposition 4.1 (Master ambiguity statistic). *Let $X_1, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} P$, and suppose the admissible sets $A_\tau(X_i)$ are available. Define*

$$\hat{D}^* := \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{|A_\tau(X_i)| \geq 2\}, \quad \varepsilon_N := \sqrt{\frac{\log(2/\delta)}{2N}}.$$

Then, with probability at least $1 - \delta$,

$$|\hat{D}^* - D_\tau^*| \leq \varepsilon_N.$$

On the same event,

$$\text{diam}_{\tau, \zeta}(f_{\text{NL}}, P) \in \left[(\hat{D}^* - \varepsilon_N)_+, \min\{1, \hat{D}^* + \varepsilon_N + 2\zeta\} \right].$$

Under fixed-description target-blind supervision, half the lower endpoint is also a valid one-sided lower confidence bound on the minimax risk floor witnessed by the maximal-spread pair.

Proof. Set $Z_i := \mathbf{1}\{|A_\tau(X_i)| \geq 2\}$. The variables $Z_1, \dots, Z_N$ are i.i.d. Bernoulli with mean D_τ^* , and $\hat{D}^* = N^{-1} \sum_i Z_i$. Hoeffding's inequality therefore gives, with probability at least $1 - \delta$,

$$(\hat{D}^* - \varepsilon_N)_+ \leq D_\tau^* \leq \min\{1, \hat{D}^* + \varepsilon_N\}.$$

On this event, substituting these bounds into $D_\tau^* \leq \text{diam}_{\tau, \zeta}(f_{\text{NL}}, P) \leq D_\tau^* + 2\zeta$ and using $\text{diam}_{\tau, \zeta}(f_{\text{NL}}, P) \leq 1$ gives the stated confidence band.

Finally, the maximal-spread pair is admissible and has disagreement D_τ^* . Under target-blind supervision its two-point minimax floor is $D_\tau^*/2$, which is at least $(\hat{D}^* - \varepsilon_N)_+/2$ on the same event. $\square$

The geometry places the diameter in the partially identified interval $[D_\tau^*, \min\{1, D_\tau^* + 2\zeta\}]$, where truncation reflects that a disagreement probability cannot exceed one (Imbens and Manski, 2004). The audit estimates D_τ^* , while ζ remains a fixed tolerance. Propagating a confidence interval for D_τ^* through these deterministic endpoints therefore yields the diameter band directly; no additional correction for an estimated, potentially collapsing identification slack is required (Stoye, 2009b).

Remark 4.2 (Sharper radii via empirical Bernstein). *We use Hoeffding's inequality for a simple distribution-free certificate. At a fixed sample size, an empirical-Bernstein bound (Maurer and Pontil, 2009) replaces the Hoeffding radius $\varepsilon_N = \sqrt{\log(2/\delta)/(2N)}$ with a variance-adaptive radius of order $\sqrt{\hat{D}^* \log(2/\delta)/N} + \log(2/\delta)/N$. This can substantially tighten the certificate when overlap is rare; the remaining arguments are unchanged after substituting any valid radius. If the auditor chooses the sample size adaptively, a fixed- N bound is insufficient. Betting intervals (Waudby-Smith and Ramdas, 2024) or confidence sequences (Howard et al., 2021) provide variance-adaptive, anytime-valid alternatives.*

4.2 The blind pairwise and finite-family floors

Target blindness is a structural property of the channel that we assume throughout; it is not estimated by the audit. Under this assumption the audit estimates only the disagreement mass of an independently selected admissible pair, which suffices to certify a risk floor. Fix $f_a, f_b \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ and write $D_{ab} = P(f_a \neq f_b)$ for their disagreement mass. For $u \in \{a, b\}$, let P_u denote the one-example observation law induced by target f_u through the fixed-description base channel. A measurable, possibly randomized learner maps a sample $S \sim P_u^{\otimes m}$ to a classifier $\hat{h}(S)$ of risk $d_P(\hat{h}(S), f_u)$. Independently of both S and the choice of (f_a, f_b) , draw $z_1, \dots, z_N \sim P$ and estimate the disagreement mass by

$$\hat{D}_{ab} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{f_a(z_i) \neq f_b(z_i)\}.$$

If the channel is target-blind on $\{f_a, f_b\}$ (so $P_a = P_b$ and $P_a^{\otimes m} = P_b^{\otimes m}$ for every m), the two-point argument of Theorem 3.2(b) gives the population floor $\max_{u \in \{a, b\}} \mathbb{E}_{S \sim P_u^{\otimes m}}[d_P(\hat{h}(S), f_u)] \geq D_{ab}/2$, uniformly in m and over all measurable learners. Its empirical version replaces D_{ab} by the one-sided Hoeffding lower bound $D_{L, ab} := [\hat{D}_{ab} - \sqrt{\log(2/\delta)/2N}]_+$, valid with probability at least $1 - \delta$, so that the certified floor is $D_{L, ab}/2$; we record both as Theorem B.3 in Section B. The certificate is algorithm-independent: $\hat{h}$ ranges over all measurable rules, so the floor is a property of the supervision channel, not of any learner. Its only sampling requirement is independence of the N audit inputs; no repeated oracle call enters. We report the two-sided radius $\varepsilon_N = \sqrt{\log(2/\delta)/2N}$ throughout under a single confidence budget, which for a one-sided floor is deliberately conservative.

Lifting the certificate from a single pair to a fixed finite family $\{f^{(1)}, \dots, f^{(K)}\} \subseteq \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ costs only a union bound over the $\binom{K}{2}$ pairs, widening each per-pair radius to $\sqrt{\log(2\binom{K}{2})/\delta}/2N$ and certifying

$$\inf_{\hat{h}} \max_k \mathbb{E}[d_P(\hat{h}(S), f^{(k)})] \geq \max_{a < b} D_{L,ab}/2$$

(Theorem B.4 in Section B). Once the admissible sets are available, the certified floor is a property of the fixed labelings and the audit inputs alone; no further reading-specific oracle calls are required.

Remark 4.3 (Scope of the minimax certificate). *Theorem B.4 bounds the worst-case risk over the retained admissible readings, a maximin (or Γ -minimax) criterion in the sense of the maximin expected-utility model for a set of priors (Gilboa and Schmeidler, 1989). It protects against every retained reading and, in particular, places no prior over them, so it is the appropriate target when no defensible distribution over intended readings is available. Should external evidence supply reading weights, one may instead evaluate the prior-weighted average risk; the present certificate concerns the worst-case quantity, not that average.*

This overlap statistic also characterizes the population zero-overlap regime and upper-bounds unresolved overlap from data.

Corollary 4.4 (Zero-overlap regime). *If $D_\tau^\star = 0$, then $\text{diam}_P(\text{Sel}_\tau) = 0$ and $\text{diam}_{\tau, \zeta}(f_{\text{NL}}, P) \leq 2\zeta$; conversely, $\text{diam}_{\tau, \zeta}(f_{\text{NL}}, P) > 2\zeta$ implies $D_\tau^\star \geq \text{diam}_{\tau, \zeta}(f_{\text{NL}}, P) - 2\zeta > 0$. Consequently, the certified upper bound $\hat{D}^\star + \varepsilon_N$ on $D_\tau^\star$ upper-bounds every pairwise $D/2$ obstruction in $\mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ by $\zeta + \frac{1}{2}(\hat{D}^\star + \varepsilon_N)$.*

Proof. Both directions follow from the representation sandwich of Theorem 2.11. If $D_\tau^\star = 0$ then $\text{diam}_P(\text{Sel}_\tau) = 0$, so $\text{diam}_{\tau, \zeta} \leq 2\zeta$; if instead $\text{diam}_{\tau, \zeta} > 2\zeta$ then $D_\tau^\star \geq \text{diam}_{\tau, \zeta} - 2\zeta > 0$. Substituting the confidence bound $\hat{D}^\star + \varepsilon_N$ of Theorem 4.1 for $D_\tau^\star$ yields the stated simultaneous upper bound on pairwise obstructions. $\square$

The statistic $D_\tau^\star$ thus gives a population characterization of ambiguity at resolution 2ζ , backed by a finite-sample upper confidence bound: an overlap certified below the resolution means the residual disagreement is coverage slack rather than substantive, while any diameter above it must register in $D_\tau^\star$.

4.3 Sampled decoding: the plug-in radius

When the decoding law is accessible only through samples, the label probabilities are not observed directly and the admissible sets must be estimated.

At each of N i.i.d. held-out inputs $z_i \sim P$ the auditor draws r i.i.d. labels $y_{i,1}, \dots, y_{i,r} \sim \pi_{\text{LLM}_\theta}(\cdot \mid f_{\text{NL}}, z_i)$, independently across inputs, and forms the empirical decoding $\hat{\pi}_i(y) = \frac{1}{r} \sum_{t=1}^r \mathbf{1}\{y_{i,t} = y\}$, the plug-in admissible set $\hat{A}_\tau(z_i) = \{y : \hat{\pi}_i(y) \geq \tau\}$, and the plug-in overlap statistic $\hat{D}_{\text{plug}}^\star = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{|\hat{A}_\tau(z_i)| \geq 2\}$.

Thresholding empirical frequencies rather than exact probabilities widens the certificate radius by two terms. A finite-depth error arises from misclassifying admissibility when a label sits near τ , and it decays with the number of decodings r . What does not decay is the mass of inputs that lie near the threshold in the first place, measured by the *threshold-margin mass*

$$\kappa(\xi) := \mathbb{P}_{x \sim P} \left[\exists y \in \mathcal{Y} : |\pi_{\text{LLM}_\theta}(y \mid f_{\text{NL}}, x) - \tau| \leq \xi \right], \quad \xi > 0,$$

the probability that some label falls within ξ of τ ; this term persists at any sampling depth.

Corollary 4.5 (Plug-in master statistic under a threshold margin). *With $C = |\mathcal{Y}|$, for each fixed $\xi > 0$ and with probability at least $1 - \delta$ over the audit,*

$$|\hat{D}_{\text{plug}}^* - D_\tau^*| \leq \varepsilon_{N,r}(\xi) := \underbrace{\sqrt{\frac{\log(2/\delta)}{2N}}}_{\text{sampling}} + \underbrace{\kappa(\xi)}_{\text{margin}} + \underbrace{2C e^{-2r\xi^2}}_{\text{depth}},$$

and $[(\hat{D}_{\text{plug}}^* - \varepsilon_{N,r}(\xi))_+, \min\{1, \hat{D}_{\text{plug}}^* + \varepsilon_{N,r}(\xi) + 2\zeta\}]$ is a two-sided confidence band for $\text{diam}_{\tau,\zeta}$.

Proof sketch; full proof in Section B. Call an input ξ -safe if every label probability lies more than ξ from τ , so the unsafe mass is $\kappa(\xi)$. At a ξ -safe input, a label-wise Hoeffding union bounds the chance of a wrong admissibility decision by $2C e^{-2r\xi^2}$; the plug-in overlap indicator can therefore differ from the population one only on the unsafe mass or through this depth error, so their means differ by at most $\kappa(\xi) + 2C e^{-2r\xi^2}$. An outer Hoeffding bound over the N i.i.d. indicators adds the sampling term $\sqrt{\log(2/\delta)/2N}$, and the triangle inequality assembles the three into $\varepsilon_{N,r}(\xi)$. Transporting the bound to $\text{diam}_{\tau,\zeta}$ adds only the deterministic 2ζ at the upper end. $\square$

The margin mass $\kappa(\xi)$ is the only term of $\varepsilon_{N,r}(\xi)$ that no sampling depth removes; the residual gap between D_τ^* and the full diameter is the identification tolerance 2ζ , fixed by the description rather than incurred by search or finite samples. A positive threshold margin is unavoidable for a uniform finite-depth certificate: labels whose probabilities lie arbitrarily close to τ cannot be classified as admissible or inadmissible with uniformly controlled error from finitely many decoding draws (Theorem B.5).

4.4 Canonical-Pair Certificate

We now assemble the preceding certificates into a single audit of the canonical maximal-spread pair (f_τ^+, f_τ^-) . This pair is a deterministic functional of $(\pi_{\text{LLM}_\theta}, \tau, \prec)$, fixed before any audit data are drawn. Since the data neither select nor modify it, its disagreement is estimated directly, with no sample splitting and no data-selection correction of the type required for adaptively chosen targets (Berk et al., 2013). Algorithm 4.1 states the procedure.

Theorem 4.6 (Certificate from empirical overlap). *Under Assumptions (A1) and (A3), fix f_{NL} , the frozen LLM_θ and template Π , a threshold $\tau \in (0, \frac{1}{2})$, a coverage tolerance $\zeta \in (0, \frac{1}{2})$, and a confidence $\delta \in (0, 1)$, and suppose the supervision channel is target-blind. From N i.i.d. held-out inputs $z_1, \dots, z_N \sim P$ with observed admissible sets $A_\tau(z_i)$, form*

$$\hat{D}^* = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{|A_\tau(z_i)| \geq 2\}, \quad \varepsilon_N = \sqrt{\frac{\log(2/\delta)}{2N}}.$$

Then, with probability at least $1 - \delta$ over the audit and simultaneously for every sample size m :

(a) (Certified blind-channel floor.) *Every learner $\hat{h}$ obeys*

$$\sup_{f \in \mathcal{F}_{\tau,\zeta}(f_{\text{NL}})} \mathbb{E}_{S \sim P_f^{\otimes m}} [d_P(\hat{h}(S), f)] \geq \frac{1}{2}(\hat{D}^* - \varepsilon_N)_+,$$

so in particular the blind-channel minimax risk over the class is at least this floor.

(b) (Diameter bracket.) *The reported floor lies within ε_N of $\frac{1}{2}D_\tau^*$, which satisfies $\frac{1}{2}D_\tau^* \geq \frac{1}{2}(\text{diam}_{\tau,\zeta} - 2\zeta)$; consequently the gap between the reported floor and the largest pairwise $D/2$ obstruction in $\mathcal{F}_{\tau,\zeta}(f_{\text{NL}})$ is at most $\zeta + \varepsilon_N$.*

Input: description f_{NL} ; frozen model LLM_θ ; prompt template Π ; threshold $\tau \in (0, \frac{1}{2})$; coverage tolerance ζ ; confidence δ ; held-out inputs $z_1, \dots, z_N \sim P$; and an access mode, either exposed probabilities or sampled decoding. Under sampled decoding, also fix a margin ξ and split the budget as $\delta = \delta_{\text{audit}} + \delta_\kappa$; supply either a prespecified deterministic upper bound on $\kappa(\xi)$ (which consumes no confidence, so $\delta_{\text{audit}} = \delta$) or one certified at level δ_κ on an independent split.

Output: a one-sided floor certificate valid with probability $\geq 1 - \delta$.

1. **Admissible sets.** For each z_i , form $A_\tau(z_i) = \{y : \pi_{\text{LLM}_\theta}(y \mid f_{\text{NL}}, z_i) \geq \tau\}$ from exposed probabilities, or the r -sample plug-in $\hat{A}_\tau(z_i)$ under sampled decoding (Theorem 4.5).
2. **Statistics.** Form the overlap $\hat{D}^* = \frac{1}{N} \sum_i \mathbf{1}\{|A_\tau(z_i)| \geq 2\}$ and the multiplicity statistic $\hat{V}^* = \frac{1}{N} \sum_i \phi(A_\tau(z_i))$, with $\phi(A) = 1 - 1/|A|$ and $\phi(\emptyset) = 0$ (Equation (2)), using $\hat{A}_\tau$ in place of A_τ under sampled decoding.
3. **Radius.** Set $\varepsilon = \sqrt{\log(2/\delta)/2N}$ under exposed probabilities, or the enlarged plug-in radius $\varepsilon = \varepsilon_{\text{audit}} + \kappa(\xi) + 2Ce^{-2r\xi^2}$ of Theorem 4.5 under sampled decoding, with $\varepsilon_{\text{audit}} = \sqrt{\log(2/\delta_{\text{audit}})/2N}$; a union bound then makes the certificate valid at the total level δ .
4. **Certificate.** Return the certified pairwise floor $\frac{1}{2}(\hat{D}^* - \varepsilon)_+$, and, for the exact core value, the certificate of Theorem 4.9 (exposed) or Theorem 4.10 (sampled). Report every value alongside the access mode, model, template, N , and (δ, τ, ζ) .

Note. The pair (f_τ^+, f_τ^-) is a deterministic functional of $(\pi_{\text{LLM}_\theta}, \tau, \prec)$; under sampled decoding the corrected statistics of step 2 estimate its population overlap, and the pair induced by the sampled sets is not treated as a population target.

Algorithm 4.1: Blind-channel floor certificate.

Proof sketch; full proof in Section B. Two-sided Hoeffding on the i.i.d. indicators $\mathbf{1}\{|A_\tau(z_i)| \geq 2\}$ gives $|\hat{D}^* - D_\tau^*| \leq \varepsilon_N$ with probability at least $1 - \delta$; work on this event. For (a), the canonical pair (f_τ^+, f_τ^-) is admissible (Theorem 2.11) and satisfies $\{f_\tau^+ \neq f_\tau^-\} = \{|A_\tau| \geq 2\}$, so its disagreement mass equals $D_\tau^* \geq \hat{D}^* - \varepsilon_N$. Target-blindness renders the two targets indistinguishable through the channel, so by the two-point floor of Theorem 3.2(b) the worst-case risk is at least half this mass, uniformly in m ; and since the pair is a deterministic functional of $(\pi_{\text{LLM}_\theta}, \tau, \prec)$, fixed independently of the audit, the whole budget δ funds the single estimate $\hat{D}^*$ without splitting (the $K = 2$ case of Theorem B.3). For (b), the representation bracket gives $D_\tau^* \geq \text{diam}_{\tau, \zeta} - 2\zeta$, while any admissible pair has floor $\frac{1}{2}d_P \leq \frac{1}{2}\text{diam}_{\tau, \zeta} \leq \frac{1}{2}D_\tau^* + \zeta$; together with $|\hat{D}^* - D_\tau^*| \leq \varepsilon_N$ this places every such floor within $\zeta + \varepsilon_N$ of the reported one. $\square$

Remark 4.7 (Scope, sampled decoding, and reading a zero certificate). *The certificate is a minimax lower bound over $\mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$: it rules out any learner that does uniformly better across the class, and requires no realizability. Should the operative target lie in $\mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$, equivalently $f(x) \in A_\tau(x)$ outside a set of P -measure below ζ , the floor applies to the class containing it; but absent further identification it does not lower-bound risk at that particular target.*

Under sampled decoding the plug-in radius of Theorem 4.5 replaces ε_N , and validity requires an upper bound on the threshold-margin mass $\kappa(\xi)$; without one the certificate is not computable from decoding samples alone. If that bound is certified at level δ_1 on an independent split and the overlap estimate holds at level δ_2 , a union bound makes the overall certificate valid at level $\delta_1 + \delta_2$.

Finally, a zero lower certificate does not establish safety or the absence of ambiguity. It can arise from genuinely small overlap at the audited resolution, from insufficient sampling or decoding depth, or from a judge that assigns negligible probability to admissible alternatives. Distinguishing these causes requires the reported uncertainty terms, the access diagnostics, and comparisons across prespecified configurations.

4.5 Exact Minimax-Value Certificate

The canonical-pair floor of Theorem 4.6 is the headline certificate; this subsection is an optional sharpening and may be skipped on a first reading. The overlap floor uses only whether each admissible set is a singleton. Its full multiplicity carries more information, and exploiting it sharpens the guarantee from a floor on worst-case risk to a certificate for the exact blind-channel value $V_\tau^\star$.

The audited admissible sets also yield the multiplicity estimator

$$\widehat{V}^\star := \frac{1}{N} \sum_{i=1}^N \left(1 - \frac{1}{k(z_i)}\right) = \frac{1}{N} \sum_{i=1}^N \phi(A_\tau(z_i))$$

of $V_\tau^\star$, where $k(z_i) = |A_\tau(z_i)|$ and

$$\phi(A) = 1 - \frac{1}{|A|} \text{ for } A \neq \emptyset, \quad \phi(\emptyset) = 0. \quad (2)$$

The exact value $V_\tau^\star$ is pinned to the overlap mass $D_\tau^\star$ from both sides, which is what lets the audit certify it.

Corollary 4.8 (Placement of the exact value). *With $C = |\mathcal{Y}|$, the exact blind-channel value satisfies*

$$\frac{1}{2}D_\tau^\star \leq V_\tau^\star \leq \left(1 - \frac{1}{C}\right)D_\tau^\star, \quad D_\tau^\star - V_\tau^\star = \mathbb{E}_x \left[\frac{\mathbf{1}\{k(x) \geq 2\}}{k(x)} \right],$$

the lower bound attaining equality if and only if $k(x) \leq 2$ for P -a.e. x . Under target-blindness on Sel_τ (Theorem 3.5), the neighborhood minimax is bracketed uniformly in m ,

$$V_\tau^\star \leq R_m(\mathcal{F}_{\tau,\zeta}) \leq V_\tau^\star + \zeta.$$

Proof. Write $k = k(x)$. The summand $1 - 1/k$ vanishes at $k = 1$ and lies in $[\frac{1}{2}, 1 - \frac{1}{C}]$ for $2 \leq k \leq C$, so $(1 - 1/k)\mathbf{1}\{k \geq 2\}$ is squeezed between $\frac{1}{2}\mathbf{1}\{k \geq 2\}$ and $(1 - \frac{1}{C})\mathbf{1}\{k \geq 2\}$; taking expectations gives the sandwich. Subtracting termwise,

$$\mathbf{1}\{k \geq 2\} - \left(1 - \frac{1}{k}\right) = \frac{\mathbf{1}\{k \geq 2\}}{k},$$

and integrating gives the gap identity; the lower bound is tight exactly when the factor $\frac{1}{2}$ is attained a.e., that is, $k(x) \leq 2$.

For the neighborhood bracket, the lower bound is immediate from $\text{Sel}_\tau \subseteq \mathcal{F}_{\tau,\zeta}(f_{\text{NL}})$. For the upper bound, every $f \in \mathcal{F}_{\tau,\zeta}(f_{\text{NL}})$ admits some $f_0 \in \text{Sel}_\tau$ with $d_P(f, f_0) < \zeta$, so the exact-value-optimal learner $\hat{h}_\star$ satisfies

$$\mathbb{E}[d_P(\hat{h}_\star, f)] \leq \mathbb{E}[d_P(\hat{h}_\star, f_0)] + d_P(f_0, f) < V_\tau^\star + \zeta;$$

taking the supremum over f gives the bracket, uniformly in m . $\square$

The overlap floor $\frac{1}{2}D_\tau^\star$ therefore equals $V_\tau^\star$ under binary overlap and lower-bounds it otherwise, while the neighborhood minimax stays within ζ of $V_\tau^\star$. Rendered finite-sample through $\widehat{D}^\star$ alone, without the multiplicity estimator, this placement already yields a two-sided interval for $V_\tau^\star$.

Concretely, for any finite admissible family $\mathcal{F}_K$ containing the maximal-spread pair, the blind-channel minimax risk $V_{\text{blind}}(\mathcal{F}_K)$ is independent of m and, with probability at least $1 - \delta$, lies in $[\frac{1}{2}(\widehat{D}^\star - \varepsilon_N)_+, \widehat{D}^\star + \varepsilon_N + 2\zeta]$; the same interval brackets $V_\tau^\star$ (Theorem B.6, proved in Section B). The certificate below sharpens its lower endpoint using the multiplicity estimator $\widehat{V}^\star$.

Proposition 4.9 (Standalone certificate for the exact blind-channel value). *Suppose the supervision channel is target-blind, and form $\hat{D}^*$ and $\hat{V}^*$ from N i.i.d. held-out inputs. With radius $\varepsilon_N = \sqrt{\log(2/\delta)/2N}$ and $C = |\mathcal{Y}|$, define*

$$C_{\text{joint}} := \max \left\{ \left(\hat{V}^* - \left(1 - \frac{1}{C}\right) \varepsilon_N \right)_+, \frac{1}{2} \left(\hat{D}^* - \varepsilon_N \right)_+ \right\}.$$

Then, with probability at least $1 - \delta$,

$$V_\tau^* \geq C_{\text{joint}} \quad \text{and} \quad R_m(\mathcal{F}_{\tau, \zeta}) \geq C_{\text{joint}} \quad \text{for every } m.$$

The second branch of C_{joint} is the certified overlap floor, so the certificate never falls below it; it is strictly larger precisely when $(\hat{V}^* - (1 - 1/C)\varepsilon_N)_+ > \frac{1}{2}(\hat{D}^* - \varepsilon_N)_+$, which, when neither branch is clipped at zero, reads $\hat{V}^* - \frac{1}{2}\hat{D}^* > (\frac{1}{2} - \frac{1}{C})\varepsilon_N$.

Proof. Allocate failure probability $\delta/2$ to each of two one-sided Hoeffding bounds. The first gives $D_\tau^* \geq \hat{D}^* - \varepsilon_N$: its one-sided radius at level $\delta/2$ is $\sqrt{\log(1/(\delta/2))/(2N)} = \varepsilon_N$, so the split incurs no additional radius. The summands $1 - 1/k(z_i)$ of $\hat{V}^*$ lie in $[0, 1 - \frac{1}{C}]$, so the second gives $V_\tau^* \geq \hat{V}^* - (1 - \frac{1}{C})\varepsilon_N$. A union bound makes both hold with probability at least $1 - \delta$; work on that event. The second bound is the first branch of C_{joint} , and $V_\tau^* \geq \frac{1}{2}D_\tau^* \geq \frac{1}{2}(\hat{D}^* - \varepsilon_N)$ is the second, so $V_\tau^* \geq C_{\text{joint}}$. Finally $\text{Sel}_\tau \subseteq \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ transfers the bound to the neighborhood minimax, $R_m(\mathcal{F}_{\tau, \zeta}) \geq C_{\text{joint}}$ for every m . $\square$

When the decoding probabilities are not exposed, the same certificate holds with each branch widened by the plug-in error $q_r(\xi)$.

Proposition 4.10 (Exact-value certificate under sampled decoding). *Suppose the admissible sets are estimated by the r -sample plug-in $\hat{A}_\tau$ at margin ξ , and form the plug-in statistics $\hat{D}_{\text{plug}}^*$ and $\hat{V}_{\text{plug}}^* = \frac{1}{N} \sum_i \phi(\hat{A}_\tau(z_i))$. With $C = |\mathcal{Y}|$, $\varepsilon_N = \sqrt{\log(2/\delta)/2N}$, and $q_r(\xi) = \kappa(\xi) + 2Ce^{-2r\xi^2}$, the certificate*

$$C_{\text{joint}}^{\text{plug}} := \max \left\{ \left(\hat{V}_{\text{plug}}^* - \left(1 - \frac{1}{C}\right)(\varepsilon_N + q_r(\xi)) \right)_+, \frac{1}{2} \left(\hat{D}_{\text{plug}}^* - \varepsilon_N - q_r(\xi) \right)_+ \right\}$$

satisfies, with probability at least $1 - \delta$,

$$V_\tau^* \geq C_{\text{joint}}^{\text{plug}} \quad \text{and} \quad R_m(\mathcal{F}_{\tau, \zeta}) \geq C_{\text{joint}}^{\text{plug}} \quad \text{for every } m.$$

Proof sketch; full proof in Section B. Replacing exact admissible sets by their r -sample plug-ins introduces a per-input error of probability at most $q_r(\xi)$. Because ϕ and $\mathbf{1}\{|\cdot| \geq 2\}$ change only when the set changes and have ranges $[0, 1 - \frac{1}{C}]$ and $[0, 1]$, this shifts the two estimator means by at most $(1 - \frac{1}{C})q_r(\xi)$ and $q_r(\xi)$. Each bias adds to the corresponding one-sided Hoeffding radius, inflating the two branches by at most the $q_r(\xi)$ terms above. The remaining steps are the $\delta/2$ split, the placement $V_\tau^* \geq \frac{1}{2}D_\tau^*$, and the neighborhood transfer; these repeat the proof of Theorem 4.9. $\square$

Remark 4.11 (Using the audit in practice). (i) Who can run it. *The correction-free audit needs the judge's admissible-label probabilities, exposed only by a logits-serving backend (open-weight or locally hosted judges); a sampling-only API instead supports the sampled-decoding audit, valid under a threshold-margin bound and sufficient decoding depth (Theorem 4.5).* (ii) Interpretation. *A certified floor directs intervention at the information structure rather than at sample size.* (iii) What to report. *Every certified quantity is configuration-relative, so a floor is meaningful only with the tuple (model, prompt template, τ -sweep, N , δ , access mode) that Algorithm 4.1 returns alongside it.*

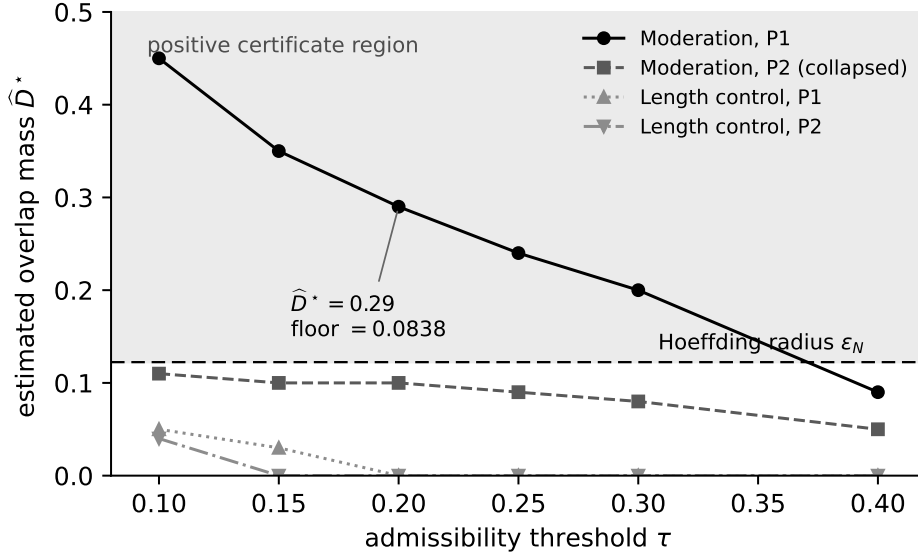


Figure 1: Correction-free exposed-probability audit at $\delta = 0.10$, $N = 100$, over the prespecified threshold sweep. The certificate is positive exactly where the estimated overlap $\hat{D}^*$ clears the Hoeffding radius ε_N ; the certified floor $\frac{1}{2}(\hat{D}^* - \varepsilon_N)_+$ is valid uniformly over all learners and sample sizes m under the fixed-description target-blind channel (Theorem 2.4), with blindness supplied by the deployment design, not established by the audit. Only the P1 moderation prompt enters the positive region; the collapsed paraphrase P2 and both exact-rule controls stay below the radius at every τ .

5 Empirical Probes

By construction, the pairwise certificate of Algorithm 4.1 is positive exactly when the observed model-induced overlap clears its finite-sample radius (the multiplicity-based exact-value certificate can be positive through a separate branch), and any positive floor it returns is sample-size independent. The empirical question is whether a real judge exhibits such overlap where the specification is genuinely underspecified and, critically, none where an exact rule leaves no room for it. We test this on one frozen judge configuration with matched exact-rule null controls, then delimit the resulting claim: does the pointwise certificate transfer to supplied candidate reading clauses, and where do implementation limits bind? These are probes, not deployment-scale benchmarks. A controlled pipeline check on constructed cases with known disagreement, a model-free ChaosNLI exact-value calculation, and all implementation details are collected in Section C.

Correction-free exposed-probability audit. We run Algorithm 4.1 in its exposed-probability mode on a frozen Qwen 2.5–3B judge, where both plug-in corrections vanish and Theorem 4.6 applies with the correction-free radius $\varepsilon_N = \sqrt{\log(2/\delta)/2N}$. The audited channel is the judge’s declared-label conditional first-token law (the class-token probabilities renormalized over the declared label set), formally defined, with all tokenization and runtime details, in Section C.10. Two prompt paraphrases (P1, P2), two exact-rule control tasks, and the threshold sweep $\tau \in \{0.10, \dots, 0.40\}$ are fixed before any model call.

Figure 1 reports the outcome across the sweep. For the moderation task under P1, $\hat{D}^* = 0.29$

clears the radius at the comparison point $\tau = 0.20$ and the audit certifies a floor of 0.0838; the certificate is positive at every swept $\tau \leq 0.30$, and for the binary label set the exact-value certificate of Theorem 4.9 coincides with this pairwise floor (numerical sweep values in Section C.8). Both exact-rule controls certify zero, as predicted: the procedure does not manufacture overlap where none exists. Paraphrase P2 also certifies zero, but for a different reason: its declared-label probability collapses onto one label (entropy diagnostic in Section C.10). Prompt sensitivity, not decoding depth, thus gates the certificate for this pair, and a zero certificate does not establish that the task is unambiguous (Theorem 4.7).

Transfer boundary. Theorem 3.9 bounds the gap between the pointwise value V_τ^* and the blind value over a supplied family of coherent readings, provided the readings track the model’s admissible geometry. We test it on two supplied candidate reading clauses for the moderation task, written independently of that geometry (they carry no independent human validation). The fit–holdout coverage estimate is favorable ($\hat{\eta}_U = 0.0000$), but the certified admissibility violation is large ($\hat{\zeta}_U = 0.4904$). Consequently, $|V_\tau^* - V_{\text{blind}}(\mathcal{C})| \leq 0.4904$ is nearly the largest bound the theorem permits. This is a direct test of the bridge, and its failure is informative because it identifies which theorem condition fails: the clauses match the aggregate mixture condition while violating the separate admissibility condition, so no semantic transfer is certified. The positive certificate above is therefore pointwise and model-relative; this is a boundary of the claim, not evidence that the judge is unambiguous. The clauses, split procedure, mixture optimization, confidence allocation, and the preregistered external-reading protocol this motivates are in Section C.9.

Implementation boundaries. Under sampling-only access at the audited breadth ($r = 3$), the finite-depth term of the plug-in radius (Theorem 4.5) saturates and every certified floor is zero; a positive floor at margin $\xi = 0.05$ would require roughly $r \approx 877$ decodings per input, which is why the principal audit exposes probabilities (design, per-configuration results, and cost–precision trade-off in Section C.5). Separately, a controlled pipeline check on constructed cases with known disagreement recovers the prescribed certificates exactly: zero where two readings never disagree, and a conservative floor of 0.166 where they disagree by $D = 0.4$ (Section C.1). Every audit fixes its design and split before any model call; source code, frozen designs, versioned outputs, and SHA-256 hashes are in the supplementary reproducibility archive (Section C.10).

6 Discussion and Limitations

Summary. NL-PAC turns the risk induced by model-admissible ambiguity into an empirically auditable quantity: for a fixed description, model, decoding threshold, and deployment distribution, it certifies a finite-sample lower bound on the worst-case risk over admissible labelings. Because this obstruction is a property of the specification–channel pair rather than of any estimator, it persists independently of the learner and its sample size.

Scope of the guarantees. The certified quantities are worst-case over admissible labelings and conditional on the sampled deployment distribution P . They characterize the frozen model–prompt–decoding configuration, not intrinsic human ambiguity, so model updates, temperature changes, or prompt edits require re-auditing. The audit requires that admissible sets be observable; under sampled decoding this means a certified threshold-margin bound and a plug-in decoding depth (Section 4.3). The blind-channel floors rely only on target blindness and the induced observation laws, not on repeated-call independence; the conditional independence of Assumption (A2) enters

only where concentration is applied to repeated oracle draws, namely the sampled-decoding analysis (Section 4.3).

The experiments establish one limited positive claim: the exposed-probability audit gives a positive model-relative certificate for the prespecified P1 prompt, with zero certificates under the exact-rule controls (Section 5). A controlled probe additionally verifies that the certificate arithmetic recovers known zero and positive disagreement cases (Section C.1).

Two negative results delimit that claim. The shallow sampled-decoding audit is vacuous at the available depth, because the finite-depth correction saturates (Section 5). The fit–holdout bridge is too loose to transfer the positive exposed-probability certificate for P1 to the supplied two-reading pool despite a zero estimated coverage gap: its admissibility bound is $\hat{\zeta}_U = 0.4904$, hence $\max\{\eta_U, \zeta_U\} = 0.4904$ (Section 5). That transfer remains open.

What the certificate licenses. A positive floor of c rules out any worst-case guarantee below c over the retained model-admissible labelings, no matter how many additional labels are collected through the same description–model–prompt channel. Lowering the floor therefore requires changing the information structure: clarifying the specification, introducing reading-revealing supervision, or changing the model or prompt. The certificate quantifies the floor but does not rank these interventions. Conversely, a zero lower certificate is not evidence of safety or of the absence of human ambiguity (Theorem 4.7).

The minimax criterion is the appropriate target when no defensible probability distribution over the retained readings is available, or when the decision rule must protect against every retained reading, as in minimax and minimax-regret decisions under set identification (Manski, 2003; Stoye, 2009a). When external evidence identifies credible reading weights, the corresponding prior-weighted risk defines a different criterion, which the present certificate does not estimate (Theorem 4.3).

This also explains why common judge diagnostics are not substitutes for the certificate. Response entropy and probability margins summarize concentration in the judge’s output distribution but do not yield a confidence-qualified lower bound on target risk. Cross-judge and judge–human disagreement measure evaluator sensitivity, yet common-mode collapse can drive these rates to zero, and they do not identify which admissible targets the deployed channel leaves unresolved. NL-PAC supplies the missing decision-theoretic step: for a prespecified channel, it maps the admissible target set to a certified minimax risk floor.

Future work. Three extensions are directly exposed by the limitations above. First, *coherent-reading coverage*: a preregistered study should elicit readings independently of model outputs, freeze the resulting clauses and a held-out input split, and then test both bridge terms (mixture coverage and admissibility) separately. A larger independently validated pool would show whether Theorem 3.9 becomes informative once held-out admissibility error is small; the frozen protocol is specified in Section C.9. Second, *cross-model and natural-distribution replication*: replication across model families and naturally occurring inputs, with a complete high-depth sampled-decoding audit, would assess the stability of the exposed-probability certificate and determine whether the threshold-margin or the finite-depth term is the binding constraint (Section 5). Third, *non-blind and intervention channels*: embedding the blind channel as the zero-sensitivity endpoint of a family of intervention channels would quantify how much reading-revealing information the channel must expose to push risk below the blind-channel floor (Theorem 3.2).

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A Notation

Table 2 collects the recurring symbols.

B Deferred Proofs and Supporting Statements

This appendix collects the proofs deferred from the main text, grouped to mirror the body and ordered within each group by appearance.

B.1 Proofs for model-admissible geometry

Proof of Theorem 2.11. ($\supseteq$). Suppose $d_P(u, v) < \zeta$ for some $v \in \text{Sel}_\tau$. Since $v(x) \in A_\tau(x)$ for P -a.e. x , the event $\{u(x) \in A_\tau(x)\}$ contains $\{u = v\}$ up to a null set, and disagreement with v has mass $d_P(u, v)$, so $\mathbb{P}_P[\pi_{\text{LLM}_\theta}(u(x) \mid f_{\text{NL}}, x) \geq \tau] = \mathbb{P}_P[u(x) \in A_\tau(x)] \geq \mathbb{P}_P[u = v] = 1 - d_P(u, v) > 1 - \zeta$, which is membership in $\mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ (Theorem 2.8). ($\subseteq$). Let $u \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ and set $G := \{x : u(x) \in A_\tau(x)\}$; G is measurable because $x \mapsto \pi_{\text{LLM}_\theta}(u(x) \mid f_{\text{NL}}, x) = \sum_{y \in \mathcal{Y}} \mathbf{1}\{u(x) = y\} \pi_{\text{LLM}_\theta}(y \mid f_{\text{NL}}, x)$ is measurable under Assumption (A1), and membership gives $\mathbb{P}_P[G] > 1 - \zeta$. Define $v := u$ on G , $v := y_{(1)}(x)$ (the top-ranked admissible label of Theorem 2.9) on $G^c \cap \{A_\tau(x) \neq \emptyset\}$, and $v := y_0$ on the remaining set $G^c \cap \{A_\tau(x) = \emptyset\}$, which is P -null by the nonemptiness hypothesis; v is then a measurable total map into $\mathcal{Y}$. Then $v \in \text{Sel}_\tau$ and $d_P(u, v) \leq \mathbb{P}_P[G^c] < \zeta$, so $d_P(u, \text{Sel}_\tau) < \zeta$. (*Diameter of the core.*) Two selections can disagree at x only if $A_\tau(x)$ contains at least two labels, so $d_P(u, v) \leq D_\tau^*$ for all $u, v \in \text{Sel}_\tau$. The two selections taking the top and a second admissible label wherever $|A_\tau(x)| \geq 2$ disagree on all of that set, so the bound is attained and $\text{diam}_P(\text{Sel}_\tau) = D_\tau^*$. (*Sandwich.*) Since $\text{Sel}_\tau \subseteq \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$, the previous step gives $D_\tau^* \leq \text{diam}_{\tau, \zeta}$. For the upper bound, take $u, v \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ with selections $\tilde{u}, \tilde{v} \in \text{Sel}_\tau$ as in ($\subseteq$); the triangle inequality for d_P gives $d_P(u, v) \leq d_P(\tilde{u}, \tilde{v}) + d_P(u, \tilde{u}) + d_P(v, \tilde{v}) < D_\tau^* + 2\zeta$, and taking the supremum yields $\text{diam}_{\tau, \zeta} \leq D_\tau^* + 2\zeta$. $\square$

Proof of Theorem 2.12. (a) The upper bound $\text{diam}_{\tau, \zeta} \leq D_\tau^* + 2\zeta$ is Theorem 2.11; it remains to match it. Fix $\varepsilon \in (0, \zeta)$. Off the overlap region $B^* = \{x : |A_\tau(x)| \geq 2\}$ the admissible set is a singleton $\{a(x)\}$, with $x \mapsto a(x)$ measurable (the top admissible label of Theorem 2.9) and nonempty P -a.e. by Assumption (A3). Since P is nonatomic on $\{x : |A_\tau(x)| = 1\}$, which has mass at least $2\zeta > 2(\zeta - \varepsilon)$, choose disjoint measurable subsets S_1, S_2 of it with $P(S_1) = P(S_2) = \zeta - \varepsilon$. Let (f_τ^+, f_τ^-) be the maximal-spread pair, and let $b(x)$ be the $\prec$ -smallest label distinct from $a(x)$ (measurable, and well defined since $|\mathcal{Y}| \geq 2$). Define

$$f_1(x) = \begin{cases} b(x) & x \in S_1, \\ f_\tau^+(x) & \text{else,} \end{cases} \quad f_2(x) = \begin{cases} b(x) & x \in S_2, \\ f_\tau^-(x) & \text{else.} \end{cases}$$

Each f_j deviates from an almost-everywhere admissible selection only on a set of mass $\zeta - \varepsilon < \zeta$, so $f_1, f_2 \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ by Theorem 2.11. Off B^* both selections equal $a(x)$, so f_1 and f_2 disagree on all of $S_1 \cup S_2$ (on each S_j exactly one takes the value $b(x) \neq a(x)$) and, disjointly, on the overlap event $\{f_\tau^+ \neq f_\tau^-\} = B^*$; hence $d_P(f_1, f_2) \geq D_\tau^* + 2(\zeta - \varepsilon)$. Letting $\varepsilon \downarrow 0$ gives $\text{diam}_{\tau, \zeta} \geq D_\tau^* + 2\zeta$, so with the upper bound the diameter equals $D_\tau^* + 2\zeta$. The supremum is not attained by any single pair, since each admissible labeling deviates on mass strictly below ζ , whence every pairwise disagreement is strictly below $D_\tau^* + 2\zeta$.

(b) Let $f' \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ with coverage complement $N = \{x : \pi_{\text{LLM}_\theta}(f'(x) \mid f_{\text{NL}}, x) < \tau\}$, so $P(N) < \zeta$. If every atom of the purely atomic P has mass at least ζ , then N contains no atom and hence $P(N) = 0$, so $f' \in \text{Sel}_\tau$. With the immediate reverse inclusion $\text{Sel}_\tau \subseteq \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$, the two classes agree up to P -null sets, and $\text{diam}_{\tau, \zeta} = \text{diam}_P(\text{Sel}_\tau) = D_\tau^*$ by Theorem 2.11. $\square$

B.2 Proofs for blind-channel floors and exact blind values

Proof of Theorem 3.2. (a) This is Le Cam’s two-point method with the error rate as semi-distance (Tsybakov, 2009). For any classifier h and any x , the indicators satisfy $\mathbf{1}[h(x) \neq f_1(x)] + \mathbf{1}[h(x) \neq f_2(x)] \geq \mathbf{1}[f_1(x) \neq f_2(x)]$, since $h(x)$ can equal at most one of two distinct labels; taking $\mathbb{E}_{x \sim P}$ gives $d_P(h, f_1) + d_P(h, f_2) \geq D$ for every fixed h . Hence, writing p_j for densities with respect to a common dominating measure (e.g. $P_1^{\otimes m} + P_2^{\otimes m}$),

$$\begin{aligned} \mathbb{E}_{P_1^{\otimes m}}[d_P(\hat{h}, f_1)] + \mathbb{E}_{P_2^{\otimes m}}[d_P(\hat{h}, f_2)] &\geq \int (d_P(\hat{h}, f_1) + d_P(\hat{h}, f_2)) \min(p_1, p_2) \\ &\geq D \int \min(p_1, p_2) = D (1 - \text{TV}(P_1^{\otimes m}, P_2^{\otimes m})). \end{aligned}$$

Since the maximum is at least the average, dividing by two yields part (a).

(b) By Equation (1), the conditional law of the decoded label $\rho(J_j, x)$ given x agrees under f_1 and f_2 for P -almost every x (off B this is immediate, since the targets share the label there). The input marginal P is common to both experiments, so the one-example observation laws coincide, $P_1 = P_2$; hence $P_1^{\otimes m} = P_2^{\otimes m}$ and $\text{TV}(P_1^{\otimes m}, P_2^{\otimes m}) = 0$ for every m . Substituting into part (a) gives $\max_j \mathbb{E}[d_P(\hat{h}, f_j)] \geq D/2$, uniformly in m . For the fixed-description base channel the hypothesis is automatic, since $J_j \sim g(\cdot \mid f_{\text{NL}}, x)$ does not depend on the target (Theorem 2.4). $\square$

Corollary B.1 (Tightness and exact minimax value of the two-point floor). *In the two-experiment construction of Theorem 3.2, the minimax mediated risk over the target pair $\{f_1, f_2\}$ is exactly the least-favorable-prior Bayes risk of the induced binary test (Wald, 1950; Ferguson, 1967; Blackwell and Girshick, 1954); because this is a finite two-action test against a two-point prior, the equality of its minimax and least-favorable-prior values is elementary and requires no general minimax theorem. For every sample size m ,*

$$\inf_{\hat{h}} \max_{j \in \{1,2\}} \mathbb{E}_{S \sim P_j^{\otimes m}}[d_P(\hat{h}(S), f_j)] = D \sup_{q \in [0,1]} \int \min(q dP_1^{\otimes m}, (1-q) dP_2^{\otimes m}).$$

Consequently:

- (a) (Blind tightness.) *In the unresolved case of Theorem 3.2(b), the minimax risk equals $D/2$ for every sample size m , attained by the data-independent learner that returns f_1 or f_2 by a fair coin flip.*
- (b) (Symmetric tightness.) *If the least-favorable prior is uniform, $q^* = \frac{1}{2}$, then the Le Cam lower bound is tight,*

$$\inf_{\hat{h}} \max_{j \in \{1,2\}} \mathbb{E}_{S \sim P_j^{\otimes m}}[d_P(\hat{h}(S), f_j)] = \frac{D}{2} (1 - \text{TV}(P_1^{\otimes m}, P_2^{\otimes m})).$$

- (c) (Total-variation sandwich.) *For general channels,*

$$\frac{D}{2} (1 - \text{TV}(P_1^{\otimes m}, P_2^{\otimes m})) \leq \inf_{\hat{h}} \max_{j \in \{1,2\}} \mathbb{E}_{S \sim P_j^{\otimes m}}[d_P(\hat{h}(S), f_j)] \leq D (1 - \text{TV}(P_1^{\otimes m}, P_2^{\otimes m})).$$

Proof of Theorem B.1. Reduction to a two-point test. Off B the two targets coincide, and any label outside $\{f_1(x), f_2(x)\}$ errs against both targets at x ; projecting a rule pointwise onto the common label off B and onto whichever of $f_1(x), f_2(x)$ it already matches on B (breaking ties arbitrarily) therefore weakly lowers $d_P(\cdot, f_1)$ and $d_P(\cdot, f_2)$ simultaneously. Hence we may restrict to rules $\hat{h}$

valued pointwise in $\{f_1(x), f_2(x)\}$. Such rules need not be global (they may follow f_1 on part of B and f_2 elsewhere), so we do *not* assert this reduction collapses onto the two functions f_1, f_2 ; instead we identify the minimax value through the least-favorable prior.

We first show that *global* rules form a sufficient class. Fix a prior with weight $q \in [0, 1]$ on f_1 , and conditional on the sample S write $q(S)$ for the posterior weight on f_1 . A Bayes rule minimizes the posterior risk $q(S) d_P(\hat{h}, f_1) + (1 - q(S)) d_P(\hat{h}, f_2)$, which pointwise on B is minimized by $f_1(x)$ when $q(S) \geq \frac{1}{2}$ and by $f_2(x)$ otherwise. This is the *same* choice at every x , since $q(S)$ does not vary with x . Hence the Bayes rule is the global test $\hat{j}: S \mapsto \{1, 2\}$ returning $f_{\hat{j}(S)}$, so global tests are sufficient even though the feasible class is larger. The problem therefore reduces to a two-action test of f_1 against f_2 ; being a finite test against a two-point prior, its minimax risk equals the least-favorable-prior Bayes risk by the elementary finite minimax identity (Wald, 1950; Ferguson, 1967), with no general minimax theorem required. Such a test errs only on B and only when $\hat{j}(S) \neq j$, so $d_P(f_{\hat{j}(S)}, f_j) = D \mathbf{1}[\hat{j}(S) \neq j]$. Hence

$$\inf_{\hat{h}} \max_{j \in \{1, 2\}} \mathbb{E}_{P_j^{\otimes m}}[d_P(\hat{h}, f_j)] = D \inf_{\hat{j}} \max_{j \in \{1, 2\}} \mathbb{P}_{P_j^{\otimes m}}[\hat{j} \neq j] = D \sup_{q \in [0, 1]} \int \min(q dP_1^{\otimes m}, (1 - q) dP_2^{\otimes m}),$$

the last equality being the minimax (least-favorable-prior) value of the Bayes test between $P_1^{\otimes m}$ and $P_2^{\otimes m}$ (Wald, 1950; Tsybakov, 2009). (a) If $P_1 = P_2$ the integrand is $\min(q, 1 - q) dP_1^{\otimes m}$, with supremum $\frac{1}{2}$ over q , so the value is $D/2$; the fair-coin learner returning f_1 or f_2 has risk $\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot D = D/2$ against either target and attains it. (b) When the least-favorable prior is uniform, evaluating at $q = \frac{1}{2}$ gives $\frac{1}{2} \int \min(dP_1^{\otimes m}, dP_2^{\otimes m}) = \frac{1}{2}(1 - \text{TV}(P_1^{\otimes m}, P_2^{\otimes m}))$, so the value is $\frac{D}{2}(1 - \text{TV})$. (c) Evaluating the supremum at $q = \frac{1}{2}$ gives the lower bound; $\min(qa, (1 - q)b) \leq \min(a, b)$ pointwise gives $\sup_q \int \min \leq \int \min(dP_1^{\otimes m}, dP_2^{\otimes m}) = 1 - \text{TV}$, the upper bound. $\square$

Remark B.2 (Average oracle accuracy is compatible with localized unresolvedness). *The unresolved floor concerns only the portion of the ambiguity the oracle cannot separate, not the full diameter, so a nonzero floor coexists with high average oracle accuracy. Call the oracle γ -approximately correct, for a fixed $\gamma \in (0, \frac{1}{2}]$, when its aggregate error ϵ_g lies a margin γ below uniform labeling, $\epsilon_g \leq 1 - \frac{1}{|\mathcal{Y}|} - \gamma$ (in the binary case $\epsilon_g \leq \frac{1}{2} - \gamma$). Writing ϵ_g^{off} for the error rate conditional on $x \notin B$, a binary oracle that errs at the chance rate $\frac{1}{2}$ on B has average error*

$$\epsilon_g = \frac{D}{2} + (1 - D) \epsilon_g^{\text{off}},$$

which satisfies the margin condition exactly when $\frac{D}{2} + (1 - D) \epsilon_g^{\text{off}} \leq \frac{1}{2} - \gamma$; for an oracle accurate off B this holds for all sufficiently small D . Thus an oracle can be γ -approximately correct on average while remaining uninformative precisely where the candidate targets disagree.

Proof of Theorem 3.5. By hypothesis the channel is target-blind on Sel_τ , so the oracle-labeled sample has the same law for every $f \in \text{Sel}_\tau$; any randomized learner therefore induces a target-independent randomization over measurable classifiers.

Let $C = |\mathcal{Y}|$ and $L = \text{lcm}(1, \dots, C)$. Enumerate the repaired set $\bar{A}_\tau(x) = \{a_1(x), \dots, a_{k(x)}(x)\}$ in the fixed order $\prec$. For each $j \leq C$, extend $a_j(x)$ by the default value y_0 on $\{x : k(x) < j\}$. The maps k and these total maps a_j are measurable because $\mathcal{Y}$ is finite and each label's admissibility event is measurable under Assumption (A1). For $i = 1, \dots, L$, define

$$f^{(i)}(x) := a_{1 + ((i-1) \bmod k(x))}(x).$$

Each $f^{(i)}$ is measurable and belongs to Sel_τ (the set on which $\bar{A}_\tau$ repairs A_τ is P -null). Since $k(x)$ divides L , a uniform choice of i makes $f^{(i)}(x)$ uniform on $\bar{A}_\tau(x)$ for every x .

Achievability (upper bound). Let $\hat{h}_\star$ ignore the sample, draw i uniformly, and output $f^{(i)}$. For every $f \in \text{Sel}_\tau$,

$$\mathbb{E}[d_P(\hat{h}_\star, f)] = \mathbb{E}_x[\mathbb{P}_i\{f^{(i)}(x) \neq f(x)\}] = \mathbb{E}_x\left[1 - \frac{1}{k(x)}\right] = V_\tau^\star,$$

where the equality is unaffected by the repaired null set. Hence the minimax is at most $V_\tau^\star$.

Lower bound. Regard $\{f^{(i)}\}_{i=1}^L$ as an indexed family (repetitions are harmless, since a supremum dominates any weighted average). Fix any randomized learner and write $H = \hat{h}(S, W)$ for its output, including its internal random seed W . Draw a uniform random index i independently of (X, S, W) by construction, and note that by target blindness the sample law of S does not depend on i ; these two facts together make $H = \hat{h}(S, W)$ independent of i , which justifies interchanging the expectation over (x, S, W) with the probability over i below. Therefore

$$\begin{aligned} \sup_{f \in \text{Sel}_\tau} \mathbb{E}[d_P(H, f)] &\geq \frac{1}{L} \sum_{i=1}^L \mathbb{E}[d_P(H, f^{(i)})] \\ &= \mathbb{E}_{x, S, W} \left[1 - \mathbb{P}_i\{f^{(i)}(x) = H(x) \mid x, H\}\right] \\ &\geq \mathbb{E}_x \left[1 - \frac{1}{k(x)}\right] = V_\tau^\star. \end{aligned}$$

Indeed, the conditional match probability is $1/k(x)$ when $H(x) \in \bar{A}_\tau(x)$ and zero otherwise. Taking the infimum over learners gives the lower bound. The two bounds coincide, and $\hat{h}_\star$ attains the value for every sample size m . $\square$

Proof of Theorem 3.9. Target blindness on $\mathcal{C}$ makes the sample law common to all $f \in \mathcal{C}$. Averaging a randomized learner over this common law produces a data-independent randomized classifier with the same risk against every target in $\mathcal{C}$. It therefore suffices to optimize over randomized classifiers.

For any distribution λ on the finite family $\mathcal{C}$, a plurality classifier minimizes the λ -average risk:

$$\inf_{\hat{h}} \mathbb{E}_{f \sim \lambda}[d_P(\hat{h}, f)] = \inf_{\hat{h}} \mathbb{E}_x[1 - \mu_x^\lambda(\hat{h}(x))] = \mathbb{E}_x[1 - \max_y \mu_x^\lambda(y)],$$

the minimizer $h^\star(x) \in \arg \max_y \mu_x^\lambda(y)$ being measurable ($\mathcal{Y}$ finite, μ_x^λ measurable in x), and randomization not helping since $1 - \mu_x^\lambda(y) \geq 1 - \max_{y'} \mu_x^\lambda(y')$ for every y .

Upper bound. Consider the data-independent randomized classifier that, at each x , draws uniformly from $\bar{A}_\tau(x)$. This kernel is measurable because $\mathcal{Y}$ is finite and $x \mapsto \bar{A}_\tau(x)$ is measurable. For $f_a \in \mathcal{C}$, let $N_a := \{x : f_a(x) \notin \bar{A}_\tau(x)\}$. Since $f_a \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$, $P(N_a) < \zeta$. On N_a^c the classifier errs with probability $1 - 1/k(x)$, while on N_a its error is at most one. Consequently,

$$\begin{aligned} \mathbb{E}[d_P(\hat{h}, f_a)] &\leq \mathbb{E}_x \left[1 - \frac{1}{k(x)}\right] + P(N_a) \\ &< V_\tau^\star + \zeta. \end{aligned}$$

Taking the supremum over a and then the infimum over learners yields $V_{\text{blind}}(\mathcal{C}) \leq V_\tau^\star + \zeta$.

Lower bound. Suppose that $\mathcal{C}$ is η -uniformly covering, and fix the covering distribution $\lambda^\star$ from Theorem 3.8. For every learner $\hat{h}$, a maximum dominates an average, $\sup_{f \in \mathcal{C}} \mathbb{E}[d_P(\hat{h}, f)] \geq \mathbb{E}_{f \sim \lambda^\star}[d_P(\hat{h}, f)]$, so taking $\inf_{\hat{h}}$ and using the display,

$$\begin{aligned} V_{\text{blind}}(\mathcal{C}) &= \inf_{\hat{h}} \sup_{f \in \mathcal{C}} \mathbb{E}[d_P(\hat{h}, f)] \\ &\geq \inf_{\hat{h}} \mathbb{E}_{f \sim \lambda^\star}[d_P(\hat{h}, f)] &&= \mathbb{E}_x[1 - \max_y \mu_x^{\lambda^\star}(y)] \\ &= V_\tau^\star - \mathbb{E}_x[\max_y \mu_x^{\lambda^\star}(y) - \frac{1}{k(x)}] \geq V_\tau^\star - \eta, \end{aligned}$$

using $V_\tau^\star = \mathbb{E}_x[1 - 1/k(x)]$ (Theorem 3.5) and the η -coverage condition. $\square$

We now record the finite-sample pairwise and finite-family floor certificates deferred from Section 4.2. Throughout, fix $f_a, f_b \in \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$ with disagreement mass $D_{ab} = P(f_a \neq f_b)$; for $u \in \{a, b\}$ let P_u be the one-example observation law induced by target f_u through the fixed-description base channel; and, independently of the learner's sample and of the choice of pair, draw $z_1, \dots, z_N \sim P$ and set $\hat{D}_{ab} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{f_a(z_i) \neq f_b(z_i)\}$.

Theorem B.3 (Finite-sample certificate for the blind-channel floor). *Fix an admissible pair (a, b) , chosen independently of the audit sample, and a confidence level $\delta \in (0, 1)$. Suppose the supervision channel is target-blind on $\{f_a, f_b\}$, that is, $P_a = P_b$, so that the two observation laws coincide, $P_a^{\otimes m} = P_b^{\otimes m}$, for every m .*

- (a) (Population certificate.) *For every m and every measurable, possibly randomized learner $\hat{h}$ receiving m examples,*

$$\max_{u \in \{a, b\}} \mathbb{E}_{S \sim P_u^{\otimes m}} [d_P(\hat{h}(S), f_u)] \geq \frac{D_{ab}}{2}. \quad (3)$$

The bound is independent of m : no amount of supervision through the blind channel can distinguish f_a from f_b , and the worst of the two risks is at least half their disagreement mass.

- (b) (Empirical certificate.) *With the audit estimator $\hat{D}_{ab}$ above, with probability at least $1 - \delta$ over the audit sample,*

$$D_{ab} \geq D_{L,ab} := \left[\hat{D}_{ab} - \sqrt{\frac{\log(2/\delta)}{2N}} \right]_+, \quad \text{so} \quad \max_{u \in \{a, b\}} \mathbb{E}[d_P(\hat{h}(S), f_u)] \geq \frac{D_{L,ab}}{2}$$

for every m .

Proof. Write $R_u := \mathbb{E}_{S \sim P_u^{\otimes m}} [d_P(\hat{h}(S), f_u)]$ for $u \in \{a, b\}$.

(a) Since $d_P(f, g) = \mathbf{1}\{f \neq g\}$ is a pseudometric on labelings, it obeys the triangle inequality $d_P(h, f_a) + d_P(h, f_b) \geq d_P(f_a, f_b)$ for every h ; for randomized $\hat{h}$ the inequality holds conditionally on its realized output and hence in expectation. The hypothesis $P_a^{\otimes m} = P_b^{\otimes m}$ lets both risks be written under the single law $P_a^{\otimes m}$, so

$$R_a + R_b = \mathbb{E}_{S \sim P_a^{\otimes m}} [d_P(\hat{h}(S), f_a) + d_P(\hat{h}(S), f_b)] \geq d_P(f_a, f_b) = D_{ab}.$$

Therefore $\max\{R_a, R_b\} \geq \frac{1}{2}(R_a + R_b) \geq D_{ab}/2$, which is Equation (3). The argument uses no property of m , so the bound holds for every m .

(b) The audit indicators $\mathbf{1}\{f_a(z_i) \neq f_b(z_i)\}$ are i.i.d. Bernoulli with mean D_{ab} , so the one-sided Hoeffding inequality gives $\mathbb{P}[\hat{D}_{ab} - D_{ab} \geq t] \leq e^{-2Nt^2}$. Setting $t = \sqrt{\log(2/\delta)/(2N)}$ yields $D_{ab} \geq D_{L,ab}$ with probability at least $1 - \delta$, and substituting this bound into part (a) certifies the floor $D_{L,ab}/2$. $\square$

Corollary B.4 (Blind finite-family floor). *Let $\{f^{(1)}, \dots, f^{(K)}\} \subseteq \mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$, $K \geq 2$, be a finite family of admissible labelings chosen independently of the audit sample, and let the supervision channel be target-blind on this family, so that the observation laws satisfy $P_a = P_b$ for every pair. For each pair $a < b$ set $\hat{D}_{ab} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{f^{(a)}(z_i) \neq f^{(b)}(z_i)\}$ on the N i.i.d. held-out audit inputs $z_1, \dots, z_N$, and define the per-pair floor*

$$D_{L,ab} := \left[\hat{D}_{ab} - \sqrt{\frac{\log(2 \binom{K}{2} / \delta)}{2N}} \right]_+.$$

Then, with probability at least $1 - \delta$ over the audit sample and for every sample size m ,

$$\inf_{\hat{h}} \max_{1 \leq k \leq K} \mathbb{E}_{S \sim P_k^{\otimes m}} [d_P(\hat{h}(S), f^{(k)})] \geq \max_{1 \leq a < b \leq K} \frac{D_{L,ab}}{2},$$

where the infimum is over all measurable, possibly randomized learners.

Proof. Fix a pair $a < b$. The audit indicators $\mathbf{1}\{f^{(a)}(z_i) \neq f^{(b)}(z_i)\}$ are i.i.d. Bernoulli with mean D_{ab} , so the one-sided Hoeffding inequality with radius $\sqrt{\log(2^{\binom{K}{2}}/\delta)/(2N)}$ gives $D_{ab} < D_{L,ab}$ with probability at most $\delta/\binom{K}{2}$. A union bound over the $\binom{K}{2}$ pairs then makes the event

$$\mathcal{E} := \{D_{ab} \geq D_{L,ab} \text{ for all } a < b\}$$

hold with probability at least $1 - \delta$.

Work on $\mathcal{E}$, and fix any learner $\hat{h}$, any m , and any pair $a < b$. Since $\{f^{(a)}, f^{(b)}\}$ is a subset of the family, retaining only these two targets can only decrease the maximum,

$$\max_{1 \leq k \leq K} \mathbb{E}[d_P(\hat{h}(S), f^{(k)})] \geq \max\{R_a, R_b\}.$$

Target-blindness on $\{f^{(a)}, f^{(b)}\}$ gives $\max\{R_a, R_b\} \geq D_{ab}/2$ by the two-point argument of Theorem B.3(a), and $D_{ab} \geq D_{L,ab}$ on $\mathcal{E}$, so the left side is at least $D_{L,ab}/2$. The bound holds for every $\hat{h}$ and every m ; taking the infimum over $\hat{h}$ and the maximum over pairs yields the claim. $\square$

B.3 Proofs for the plug-in overlap statistic

We first record why the threshold margin cannot be removed, then prove Theorem 4.5.

Remark B.5 (Necessity of the threshold margin). *Without a threshold margin the plug-in statistic is not certifiable at any finite depth. Deciding whether $\pi_{\text{LLM}_\theta}(y \mid f_{\text{NL}}, x)$ lies above or below τ at an input where it sits within ξ of the threshold requires $\Omega(\xi^{-2})$ draws. Thus no finite depth uniformly resolves inputs arbitrarily close to τ . The term $\kappa(\xi)$ bounds the deployment mass of these inputs. The audit must fix ξ in advance; optimizing it over a data-dependent grid would require a union bound over the grid. When the API exposes the decoding probabilities exactly, r is irrelevant and Theorem 4.1 applies as stated. The margin mass $\kappa(\xi)$ is the analogue, at the admissibility threshold τ , of a Mammen–Tsybakov low-noise/margin quantity (Mammen and Tsybakov, 1999): it measures the deployment mass sitting where the plug-in decision is statistically hard, and the same margin geometry governs the fast rates available for plug-in classifiers (Audibert and Tsybakov, 2007).*

Proof of Theorem 4.5. Fix $\xi > 0$. Call an input x ξ -safe if every label satisfies $|\pi_{\text{LLM}_\theta}(y \mid f_{\text{NL}}, x) - \tau| > \xi$; by definition the unsafe inputs carry mass $\kappa(\xi)$.

Step 1 (per-input decision error). At a ξ -safe input, each empirical frequency $\hat{\pi}_i(y)$ concentrates around $\pi_{\text{LLM}_\theta}(y \mid f_{\text{NL}}, x)$, so a two-sided Hoeffding bound over the r decoding draws, taken in union across the $C = |\mathcal{Y}|$ labels, gives $\mathbb{P}[\hat{A}_\tau(x) \neq A_\tau(x)] \leq 2Ce^{-2r\xi^2}$; on the complementary event the plug-in overlap indicator agrees with the population one. For the joint draw of $z \sim P$ and its decoding samples, the indicators therefore disagree only on the unsafe mass or through this depth error,

$$\mathbb{P}[\mathbf{1}\{|\hat{A}_\tau(z)| \geq 2\} \neq \mathbf{1}\{|A_\tau(z)| \geq 2\}] \leq \kappa(\xi) + 2Ce^{-2r\xi^2},$$

so their means satisfy $|\mathbb{E}[\hat{D}_{\text{plug}}^*] - D_\tau^*| \leq \kappa(\xi) + 2Ce^{-2r\xi^2}$.

Step 2 (aggregation). The N plug-in indicators are i.i.d. $\{0, 1\}$ -valued, each a function of the independent pair (input, decoding draws), so a two-sided Hoeffding bound gives $|\hat{D}_{\text{plug}}^* - \mathbb{E}[\hat{D}_{\text{plug}}^*]| \leq$

$\sqrt{\log(2/\delta)/2N}$ with probability at least $1 - \delta$. Combining the two steps by the triangle inequality yields the stated deviation bound $|\hat{D}_{\text{plug}}^* - D_\tau^*| \leq \varepsilon_{N,r}(\xi)$, and transporting it to $\text{diam}_{\tau,\zeta}$ through Theorem 2.11 adds only the deterministic 2ζ at the upper end, giving the confidence band.

Step 3 (depth prescription). When $\kappa(\xi) > 0$, taking $r \geq \ln(2C/\kappa(\xi))/(2\xi^2)$ forces $2Ce^{-2r\xi^2} \leq \kappa(\xi)$, so the band inflates over the observed- A_τ band of Theorem 4.1 by at most $2\kappa(\xi)$; when $\kappa(\xi) = 0$, taking $r \geq \ln(2C/b)/(2\xi^2)$ makes the depth term at most any fixed budget $b > 0$. $\square$

B.4 Proofs for the end-to-end audit

Proof of Theorem 4.6. Two-sided Hoeffding on the i.i.d. indicators $\mathbf{1}\{|A_\tau(z_i)| \geq 2\}$ gives $|\hat{D}^* - D_\tau^*| \leq \varepsilon_N$ with probability at least $1 - \delta$. We argue on this event.

(a) The maximal-spread pair (f_τ^+, f_τ^-) consists of almost-everywhere admissible selections, so both members lie in $\mathcal{F}_{\tau,\zeta}(f_{\text{NL}})$ (Theorem 2.11). As a deterministic functional of $(\pi_{\text{LLM}_\theta}, \tau, \prec)$, the pair is fixed independently of the audit inputs, so no sample splitting is required and the full budget δ funds the single estimate $\hat{D}^*$; this is the $K = 2$ case of Theorem B.3. Since $\{f_\tau^+(z) \neq f_\tau^-(z)\} = \{|A_\tau(z)| \geq 2\}$, the pair's disagreement mass is $D_{+-} = D_\tau^* \geq \hat{D}^* - \varepsilon_N$. Target-blindness (Theorem 2.4) makes the two observation laws coincide, so by the two-point floor of Theorem 3.2(b) the worst-case risk is at least $D_{+-}/2 \geq \frac{1}{2}(\hat{D}^* - \varepsilon_N)$, uniformly in m ; since risk is nonnegative, it is therefore at least the positive part $\frac{1}{2}(\hat{D}^* - \varepsilon_N)_+$ reported by the theorem.

(b) The representation bracket (Theorem 2.11) gives $D_\tau^* \geq \text{diam}_{\tau,\zeta} - 2\zeta$. Since the reported floor $\frac{1}{2}(\hat{D}^* - \varepsilon_N)_+$ is at least the untruncated quantity $\frac{1}{2}(\hat{D}^* - \varepsilon_N)$, it suffices to bound the latter:

$$\frac{1}{2}(\hat{D}^* - \varepsilon_N) \geq \frac{1}{2}D_\tau^* - \varepsilon_N \geq \frac{1}{2}(\text{diam}_{\tau,\zeta} - 2\zeta) - \varepsilon_N.$$

In the other direction, any pair $f_1, f_2 \in \mathcal{F}_{\tau,\zeta}(f_{\text{NL}})$ has floor $\frac{1}{2}d_P(f_1, f_2) \leq \frac{1}{2}\text{diam}_{\tau,\zeta} \leq \frac{1}{2}D_\tau^* + \zeta$, which exceeds the reported floor by at most $\zeta + \varepsilon_N$. This is the diameter-bracket consequence of part (b); it bounds every admissible pair's floor at once and involves no data-dependent search over pairs. $\square$

Corollary B.6 (Two-sided certified interval for the blind-channel value). *Let $\mathcal{F}_K \subseteq \mathcal{F}_{\tau,\zeta}(f_{\text{NL}})$ be a finite admissible family containing the maximal-spread pair, with blind-channel minimax risk $V_{\text{blind}}(\mathcal{F}_K)$. Then $V_{\text{blind}}(\mathcal{F}_K)$ is independent of m and*

$$\frac{1}{2}D_\tau^* \leq V_{\text{blind}}(\mathcal{F}_K) \leq D_\tau^* + 2\zeta;$$

consequently, with probability at least $1 - \delta$,

$$\frac{1}{2}(\hat{D}^* - \varepsilon_N)_+ \leq V_{\text{blind}}(\mathcal{F}_K) \leq \hat{D}^* + \varepsilon_N + 2\zeta.$$

Proof. Write $D := \text{diam}_P(\mathcal{F}_K)$. Target-blindness makes the observation law the same for every target in $\mathcal{F}_K$, so $V_{\text{blind}}(\mathcal{F}_K)$ is independent of m ; the two-point floor gives $V_{\text{blind}}(\mathcal{F}_K) \geq \frac{1}{2}D$, while the data-free learner returning f_τ^+ has risk at most D . Hence $\frac{1}{2}D \leq V_{\text{blind}}(\mathcal{F}_K) \leq D$. The maximal-spread pair lies in $\mathcal{F}_K$, so $D \geq d_P(f_\tau^+, f_\tau^-) = D_\tau^*$, and the representation bracket gives $D \leq D_\tau^* + 2\zeta$; combining with the previous display yields the population interval $\frac{1}{2}D_\tau^* \leq V_{\text{blind}}(\mathcal{F}_K) \leq D_\tau^* + 2\zeta$. The empirical interval follows by substituting $|D_\tau^* - \hat{D}^*| \leq \varepsilon_N$, which holds with probability at least $1 - \delta$. $\square$

Proof of Theorem 4.10. At each input the plug-in and population admissible sets differ with probability at most $q_r(\xi)$, by the argument proving Theorem 4.5. Both $\phi(A) = 1 - 1/|A|$ (with $\phi(\emptyset) = 0$),

of range $[0, 1 - \frac{1}{C}]$, and $\mathbf{1}\{|A| \geq 2\}$, of range $[0, 1]$, change only when the set changes, so the plug-in means are biased by

$$|\mathbb{E}\phi(\hat{A}_\tau(Z)) - V_\tau^*| \leq (1 - \frac{1}{C})q_r(\xi), \quad |\mathbb{E}\mathbf{1}\{|\hat{A}_\tau(Z)| \geq 2\} - D_\tau^*| \leq q_r(\xi).$$

The per-input decoding experiments are independent, so the plug-in summands are i.i.d.; one-sided Hoeffding bounds at level $\delta/2$ each, combined with these biases, give

$$V_\tau^* \geq \hat{V}_{\text{plug}}^* - (1 - \frac{1}{C})(\varepsilon_N + q_r(\xi)), \quad D_\tau^* \geq \hat{D}_{\text{plug}}^* - \varepsilon_N - q_r(\xi),$$

simultaneously with probability at least $1 - \delta$ by a union bound. On this event $V_\tau^* \geq \frac{1}{2}D_\tau^*$ (Theorem 4.8) dominates the second branch and the first display the first, so $V_\tau^* \geq C_{\text{joint}}^{\text{plug}}$ after taking positive parts and the maximum; the same corollary transfers the bound to $R_m(\mathcal{F}_{\tau,\zeta})$. $\square$

C Additional Experimental Diagnostics

This appendix collects diagnostics that support interpretation and reproducibility but do not enter the certified floors reported in Section 5: the controlled pipeline check, a threshold-sensitivity example, the model-free ChaosNLI check and its complete table, the sampled-decoding audit design and cost-precision trade-off, the margin-mass estimation procedure, the per-configuration Qwen results, the exposed-probability threshold sweep, the coherent-reading bridge audit design and results, and the exposed-probability channel definition and audit metadata.

C.1 Controlled pipeline check

This probe checks the certificate arithmetic on constructed cases; it does not validate the model-induced admissibility construct. On 6,000 binary inputs we fix two pairs of known disagreement: a divisibility control whose two readings never disagree ($D = 0$), and a pedagogy pair with two coherent readings (immediate performance versus long-term retention) disagreeing by $D = 0.4$. On the fixed-description target-blind channel the observed law is independent of the operative reading (Theorem 2.4), so the certified floor is $D_L/2$ (Theorem B.3; $N = 400$, $\delta = 0.10$). Table 3 recovers the prescribed outcome: a zero certificate for the control and a conservative positive floor of 0.166 for the pedagogy pair, valid at every learner sample size m . A threshold sweep on a related controlled construction follows in Section C.2.

C.2 Threshold sensitivity

Because the overlap mass $D_\tau^* = \mathbb{P}_x[|A_\tau(x)| \geq 2]$ is a threshold functional of the decoding law, the certified floor depends on the admissibility threshold τ , and the two should be reported together. On a controlled population in which a 0.40 fraction of inputs are two-way ambiguous by construction (decoding law $(0.45, 0.45, 0.10)$) and the rest near-confident $(0.85, 0.10, 0.05)$, sweeping τ at $N = 4000$, $\delta = 0.10$ yields a stable plateau: the certified floor is 0.49 for $\tau \leq 0.10$ (where even the near-confident second label clears the threshold and the count over-reads), holds at 0.19 across $\tau \in [0.12, 0.44]$ (where it tracks exactly the constructed ambiguous mass of 0.40), and falls to 0 for $\tau \geq 0.50$ (where no second label clears). The body’s choice $\tau = 0.20$ lies in this plateau. The construction illustrates the threshold dependence established in the body; it does not imply that empirical decoding laws generally exhibit a comparable plateau.

C.3 Model-free arithmetic check on ChaosNLI

As a check against ambiguity documented independently of any model, we use ChaosNLI (Nie et al., 2020), which re-annotates SNLI and MNLI with ≈ 100 human labels per item, designed to characterize persistent human disagreement (Pavlick and Kwiatkowski, 2019; Plank, 2022). We form a human-label analogue $q_{\text{human}}(\cdot | x)$ from each item’s empirical label distribution and apply the exact-value functional to it in place of π_{LLM_θ} ; this exercises the certificate arithmetic on documented human disagreement rather than a model-induced estimand. Under q_{human} the overlap statistic and human entropy agree in ordering: MNLI is more ambiguous than SNLI. The three-label distributions also exercise the $k \geq 3$ branch of the exact-value certificate, absent in binary problems; at $\delta = 0.10$ it raises the certified floor by 0.005 on MNLI and leaves it unchanged on SNLI. This model-free analogue is conditional on the published label distributions; it neither validates the model-relative admissibility construct nor constitutes a deployment study.

C.4 ChaosNLI multiclass check: full table

Table 4 gives the complete multiclass calculation, computed from the public ≈ 100 -annotation ChaosNLI label distributions (Nie et al., 2020) at the prespecified $\tau = 0.20$, $\delta = 0.10$. Three labels are admissible on 1.3% of SNLI items and 5.9% of MNLI-matched items, yielding $V_\tau^* = 0.263$ and 0.386, respectively. The corresponding half-overlap values are 0.261 and 0.376. At confidence $1 - \delta = 0.90$, the exact-value branch certifies 0.242 on SNLI and 0.366 on MNLI, while the half-overlap branch certifies 0.245 and 0.361. Sampling uncertainty therefore favors the half-overlap branch on SNLI, whereas the greater three-way admissibility on MNLI makes the exact-value certificate larger by 0.005.

C.5 Sampled-decoding audit: design and cost

This appendix gives the full design of the sampled-decoding depth-barrier diagnostic summarized in Section 5. The audit fixes Qwen 2.5–3B, two prompt paraphrases, $\tau = 0.2$, and $\delta = 0.1$. Each paraphrase is evaluated under three decoding seeds, giving 6 configurations per task. In each, the model receives the description and input but neither candidate reading; we draw $r = 3$ labels at each of $N = 100$ audit inputs and set

$$\hat{A}_\tau(x) = \{y : \hat{\pi}(y | f_{\text{NL}}, x) \geq \tau\}.$$

The model’s sampled decoding law thus determines admissibility: the experiment scores no researcher-supplied label set.

Task. The candidate-positive task is borderline content moderation under two prespecified coherent readings (a narrow one limited to explicit threats, slurs, or targeted abuse, and a broader one covering likely escalation of personal harm), and the control is the exact rule “LONG iff length exceeds 10.”

Sampling design. The $N + N_\kappa = 100 + 100$ moderation inputs are programmatically constructed borderline scenarios. The split was fixed independently of model outputs and subsequent analysis choices: a recorded design seed partitions the declared finite input population into disjoint audit and margin subsets before any model call. Because the inputs are drawn without replacement from this finite population, the Hoeffding radius stays valid, and is indeed conservative (Hoeffding, 1963; Bardenet and Maillard, 2015, §6), and every estimand is read as a finite-population mean over the synthetic, deliberately borderline distribution, whose overlap mass plausibly exceeds that of natural moderation traffic. Exact per-label Clopper–Pearson intervals on the r decodings at each of the disjoint N_κ margin inputs give the margin bound $\kappa_U = 1.0$, constructed in Section C.6.

Confidence allocation. Following Theorem 4.7, the confidence budget $\delta = 0.1$ splits as $\delta_1 = \delta_2 = \delta/2$ between the margin-mass bound and the overlap estimate, and the reported certificate is valid at their union δ .

The certified floor is zero in every configuration (Table 5) because at $r = 3$ the finite-depth term $2Ce^{-2r\xi^2}$ of the radius in Theorem 4.5 caps at 1; these zeros are the vacuous-correction case of Theorem 4.7, not evidence that the task or judge is benign. A positive floor at $\xi = 0.05$ requires $r \geq \ln(2C/b)/(2\xi^2)$ decodings for depth budget b , about $r \approx 877$ at $b = 0.05$, and the margin estimator is depth-limited in the same way. The audit does yield observed plug-in overlap values: $\hat{D}^*$ clears $\varepsilon_N = 0.1358$ in 2 of the 6 moderation configurations, while no control configuration does; the positive cells appear in Table 6. These are prompt-sensitive point estimates, not certificates; the per-configuration cells, the constant-output check, and the paraphrase-collapse pattern appear in Section C.7.

At $N = 100$ the sampling slack $\varepsilon_N = 0.1358$ prevents the audit from resolving overlap masses below roughly one eighth of the deployment distribution, regardless of decoding depth: under the confidence split (Theorem 4.7), slack ε requires $N \geq \log(4/\delta)/(2\varepsilon^2)$, so $\varepsilon = 0.05$ needs about 738 inputs. Figure 2 summarizes the resulting cost–precision trade-off: for each total call budget Q it plots the smallest certificate radius achievable by any breadth–depth split, alongside the exposed-probability radius, which needs one call per input. The shallow audit has a capped, hence vacuous, radius at $Q = 300$, whereas the exposed-probability audit reaches radius 0.1224 with $Q = 100$ calls; matching it by sampled decoding at $\xi = 0.05$ takes roughly three orders of magnitude more calls. Exposed admissible-label probabilities remove both plug-in corrections (Theorem 4.1), giving the more call-efficient access mode for open-weight judges with logits-exposing backends.

C.6 Margin-mass estimator

The margin bound κ_U reported in Section 5 is estimated on the disjoint $N_\kappa = 100$ margin inputs, with the confidence budget split between it and the overlap estimate. The audit implementation bounds the margin event using exact per-label Clopper–Pearson intervals. An input is classified as potentially within the margin if any label-probability interval intersects $[\tau - \xi, \tau + \xi]$. A union bound allocates the noncoverage budget across labels. Alternatively, the auditor may supply an externally justified upper bound on $\kappa(\xi)$. At $r = 3$, the exact three-draw intervals classify nearly every input as potentially within the margin, yielding the near-vacuous upper bound $\kappa_U = 1.0$ reported in the body. The partial high-depth cache is incomplete and therefore cannot support a reproducible margin-mass estimate. Determining whether the margin or finite-depth term dominates at certificate-valid depth requires a completed high-depth run.

C.7 Per-configuration Qwen audit results

Table 6 reports every cell of the two-prompt, three-seed Qwen 2.5–3B audit of Section 5: all six moderation configurations and the six matched length-control configurations, with certificates recomputed under the full radius of Theorem 4.5. As elsewhere, the finite-depth correction is vacuous at $r = 3$ (Section 5), so the informative columns are the plug-in statistics rather than the (zero) certified floor. The configurations show two patterns. First, moderation produces large plug-in overlap under P1 with seeds 0 and 2. P1 with seed 1 has overlap 0.01, while P2 produces constant output under all three seeds (majority fraction 1.00). The estimated admissible sets are therefore sensitive to both prompt paraphrase and shallow-decoding seed. Second, the control has small nonzero plug-in overlap in five of six configurations, with $\hat{D}^* \leq 0.08$, and never approaches the two large moderation signals. These control values are consistent with finite-decoding noise at

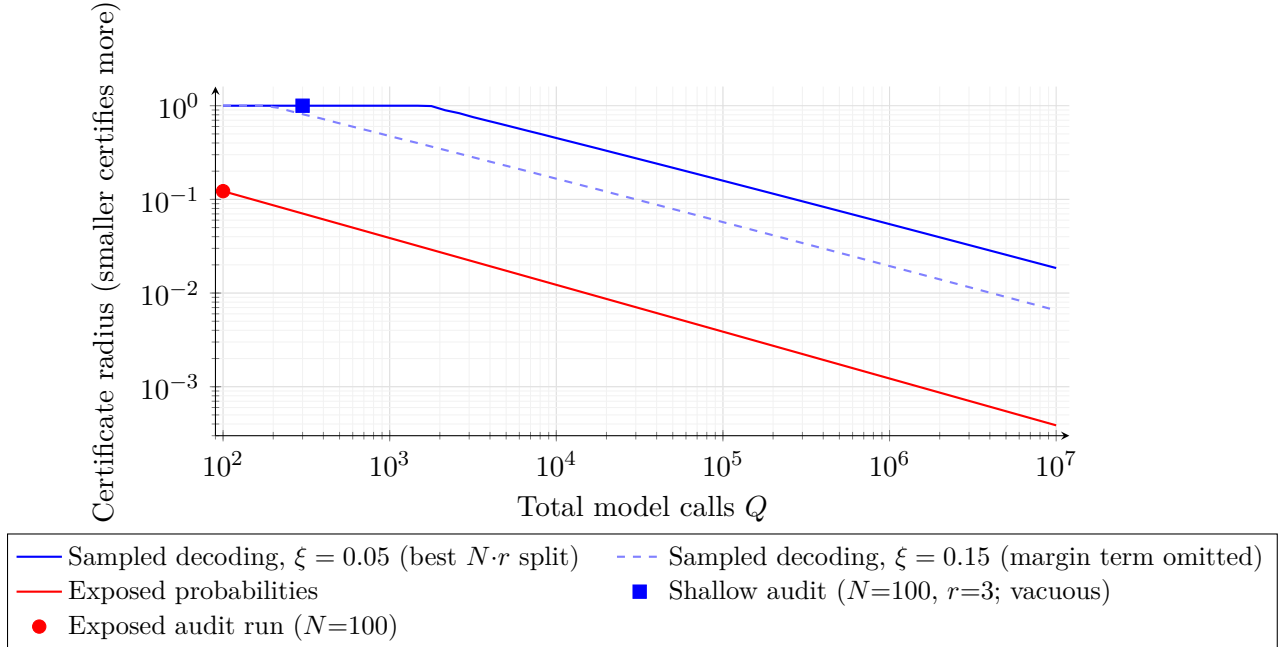


Figure 2: Cost–precision trade-off of the audit at $\delta = 0.10$ for a binary label set ($C = 2$). For each total call budget Q the sampled-decoding curves plot the smallest radius of Theorem 4.5 over breadth–depth splits; the margin term $\kappa(\xi)$ and the disjoint margin-split calls are task-dependent and omitted, so these curves are *optimistic* for sampled decoding. The exposed-probability curve plots the correction-free radius of Theorem 4.1, which needs one call per input. Markers show the two audits actually run in this section.

$r = 3$. At $r = 3$ decodings per input, per-cell plug-in probabilities are coarse (multiples of $1/3$), so the seed-to-seed variation within P1 may largely reflect Monte Carlo resolution rather than decoding-law instability.

C.8 Exposed-probability threshold sweep

The exposed-probability audit of Section 5 plots the full sweep in Figure 1 and reports its correction-free certified floor at the comparison point $\tau = 0.20$; this subsection records the numerical values. The probabilities swept here are the declared-label conditional probabilities obtained by renormalizing the raw next-token log-probabilities over the prespecified class tokens, so every floor below is relative to that constrained first-token channel (Section 5), not an unrestricted continuation channel. The full prespecified sweep for the moderation prompt P1 is monotone decreasing in τ . Across $\tau \in \{0.10, 0.15, 0.20, 0.25, 0.30, 0.40\}$ the P1 floor is 0.1638, 0.1138, 0.0838, 0.0588, 0.0388, and 0, respectively. At $N = 100$ the Hoeffding radius is $\varepsilon_N \approx 0.12$, so the positive floors arise only where $\hat{D}^*$ clears that margin; the floor vanishes once τ is large enough that fewer inputs place both class tokens above threshold.

C.9 Coherent-reading bridge audit: design and results

This subsection gives the full design behind the transfer boundary of Section 5, which tests the two conditions of Theorem 3.9 separately on the moderation task.

Readings. Two policy clauses are prespecified. The *narrow* reading: “Remove a comment only when it contains an explicit threat, slur, or targeted abusive statement. Allow it otherwise.” The *broad* reading: “Remove a comment when it is likely to escalate personal harm, including veiled or implicit targeted harassment even without explicit prohibited words. Allow it otherwise.” Each clause is inserted into a fixed reading template (“Apply this single policy reading consistently to every input: ...”), and the induced labeling $f_a(x)$ is the argmax of the exposed declared-label law under that clause. The base admissible sets $A_\tau(x)$ come from the frozen exposed-probability P1 audit of Section 5 at $\tau = 0.20$.

Split and optimization. A design-seeded permutation of the declared input population assigns 100 fit and 100 held-out inputs; the fit inputs are verified (by hash) to coincide with the frozen P1 audit inputs. The mixture weight is fitted on the fit split alone by a grid search over $\lambda \in \{0, 10^{-4}, \dots, 1\}$ minimizing the empirical coverage gap $\mathbb{E}_x[\max_y \mu_x^\lambda(y) - 1/k(x)]$, yielding $\hat{\lambda}_{\text{narrow}} = 0.50$. All reported quantities are then evaluated on the held-out split only.

Confidence allocation. The budget $\delta = 0.10$ funds three one-sided Hoeffding bounds (one for the coverage gap and one per-reading admissibility violation) via the shared radius $\sqrt{\log(3/\delta)/(2 \cdot 100)} \approx 0.1304$.

Results. The two conditions of the bridge come apart. The held-out coverage gap is -0.28 , giving the clipped upper bound $\hat{\eta}_U = 0.0000$: the fitted mixture *satisfies* the η -uniform-coverage condition (Theorem 3.8), helped by the large held-out reading disagreement (0.76). The admissibility condition fails: the held-out violation rates are 0.24 (narrow) and 0.36 (broad), giving $\hat{\zeta}_U = 0.4904$ and hence the nearly vacuous gap bound $|V_\tau^* - V_{\text{blind}}(\mathcal{C})| \leq 0.4904$. The obstruction is thus not mixture coverage but that each clause, applied globally, frequently labels outside the model’s admissible set; the clauses were written independently of the model’s admissible geometry and carry no independent human validation.

Preregistered external-reading validation (next-step design). The protocol this result motivates is fixed as follows. (i) Elicit a pool of $K \geq 5$ candidate policy readings from annotators or policy documents, independently of any model output. (ii) Freeze the clauses, the prompt template, the split sizes, τ , δ , and the analysis code, and register hashes before any model call. (iii) On the fit split, fit the mixture weights; on the held-out split, report $\hat{\eta}_U$ and per-reading $\hat{\zeta}_U$ at radius $\sqrt{\log((K+1)/\delta)/(2N_{\text{holdout}})}$. (iv) The bridge is informative iff both bounds are small; readings with large violation rates are reported, not discarded post hoc. This design tests whether Theorem 3.9 becomes informative once held-out admissibility error is small, as anticipated in the discussion.

C.10 Exposed-probability channel and audit metadata

The exposed-probability audit of Section 5 serves a frozen Qwen 2.5–3B (llama.cpp b9870) and instantiates the verbalizer V of Section 2.1 as the *constrained* first-token label kernel: $V(\cdot | s)$ places all mass on the label whose class token leads s , conditioned on that token lying in the declared class-token set $\{v_y\}_{y \in \mathcal{Y}}$. Under this V the admissibility channel is exactly the constrained first-token law

$$\pi_{\text{LLM}_\theta}(y | f_{\text{NL}}, x) = \frac{\mathbb{P}(T_1 = v_y | u)}{\sum_{y' \in \mathcal{Y}} \mathbb{P}(T_1 = v_{y'} | u)}, \quad u = \Pi(f_{\text{NL}}, x),$$

i.e. the raw class-token probabilities renormalized over the declared label set, not an unrestricted continuation probability. The certificate is stated relative to this declared-label channel and needs neither the finite-depth nor the threshold-margin correction.

As a collapse diagnostic, not itself a certified quantity, we report the mean binary entropy (nats) of the declared-label law: 0.291 for paraphrase P1 against 0.087 for P2. The order-of-magnitude

gap shows that P2’s zero certificate arises from the declared-label probability concentrating on one label (*prompt collapse*), not from small estimated overlap at comparable spread.

Each model-relative audit fixes its design and split before any model call and records the seeds, prompts, per-input probability vectors, label-token identifiers, model hash, and platform and runtime revision. The recorded vectors are the raw next-token class-token probabilities renormalized over the declared label set (the constrained first-token channel above), not full-vocabulary softmax values. The supplementary reproducibility archive provides the source code, frozen designs, versioned outputs, and SHA-256 hashes for the exposed-probability and bridge result artifacts.

D Further Related Work

Decision-theoretic status of the exact blind-channel value. Once the supervision experiment is target-independent, the reduction to a randomized rule and a least-favorable prior is classical (Wald, 1950; Blackwell, 1951; Blackwell and Girshick, 1954; Ferguson, 1967; Sion, 1958). Modern adversarial and minimax-risk classifiers solve the analogous robust 0–1 decision problem for broad uncertainty sets of joint distributions (Fathony et al., 2016; Mazuelas et al., 2023). Most notably, the adversarial zero-one loss of Fathony et al. (2016, Theorem 1, Eq. (4)) is the maximum over nonempty label subsets S of hyperplanes whose constant term is $(|S| - 1)/|S|$. The same expression arises here as the value of the finite zero-one game on $A_\tau(x)$, namely $1 - 1/|A_\tau(x)|$. Candidate-label learning is closer to our pointwise set-valued object: partial-label learning asks when training candidate sets can be disambiguated (Cour et al., 2011), but it is not formulated as the target-blind, sample-flat selector-class identity used here. The NL-PAC-specific step is the cyclic finite family whose marginal at each x is uniform on $A_\tau(x)$; it exposes the least-favorable value as $\mathbb{E}[1 - 1/|A_\tau(X)|]$ and makes that value estimable from the same held-out admissibility audit.

Superset learning and conformal prediction sets. The admissible set $A_\tau(x)$ is formally a candidate-label (superset) set, so the superset/imprecise-label line is a near neighbor of our selection-class view. Beyond the average-loss formulation of Cour et al. (2011), the generalized-loss and data-disambiguation program of Hüllermeier (2014) learns by minimizing a loss over the set of observation-consistent labelings, the same set-valued object we analyze. Its aim is to recover a single sharp model, whereas NL-PAC asks whether the mediated channel can distinguish the set’s members at all. A superficially similar set, the conformal prediction set, is a threshold set of a score function and hence formally resembles $A_\tau(x)$ (Vovk et al., 2005; Angelopoulos and Bates, 2023); the two are conceptually opposite, however. A conformal set is a property of the *predictor* and drives a coverage guarantee, while $A_\tau(x)$ is a property of the *supervision channel* and drives an identification floor: the former certifies that the truth is contained, the latter certifies that the truth cannot be pinned down.

Relation to non-identifiable noise and partial monitoring. The blind-channel floor (Theorem 3.2) is an LLM-mediated instantiation of information-theoretic noise-floor lower bounds, such as those in the malicious-error (Kearns and Li, 1993) and nasty-noise (Bshouty et al., 2002) models. However, while those floors are budgeted (vanishing as the corruption rate $\eta \rightarrow 0$), the NL-PAC floor is set by the indistinguishability of two admissible readings under the oracle channel (Equation (1)).

Instance-dependent noise (IDN) similarly makes identifiability depend on structural assumptions: bounded instance- and label-dependent noise can be learned under additional conditions (Cheng et al., 2020), whereas the unrestricted target-noise decomposition is not identified by labels alone. The systematic component of the NL-PAC oracle error is a structured instance of IDN where the

noise is generated by an LLM interpreting an ambiguous prompt. This also distinguishes NL-PAC from weak-supervision frameworks like Snorkel (Ratner et al., 2017), which aggregate multiple heuristics by estimating their accuracies, whereas we analyze whether the labeling channel can distinguish the admissible readings at all.

A supervision-channel separation between direct and mediated labels (a natural follow-up outside this paper’s scope) has a direct precedent in partial monitoring, where "hopeless" games with linear regret arise when feedback cannot separate target outcomes (Bartók et al., 2011); NL-PAC transfers this logic to an offline specification setting.

Harm specification and LMaaS opacity. The blind-channel floor has an independent information-theoretic counterpart on the alignment side: Young (2025) argues that no specification I can pin down an external target O if the conditional entropy $H(O | I) > 0$, so an information gap forces a residual error. Our ambiguity diameter $\text{diam}_{\tau, \zeta}(f_{\text{NL}}, P) > 0$ is the operational witness of this gap, and the master statistic (Theorem 4.1) estimates it from data. Judge-mediated supervision is increasingly delivered through proprietary, API-only models, the Language-Models-as-a-Service (LMaaS) paradigm, whose opacity is documented as a systematic obstacle to evaluation, reproducibility, reliability, and trustworthiness (La Malfa et al., 2024); the operative reading is then often unobservable by construction, which is exactly the reading-blind regime the floor and its audit are designed to certify.

Rating indeterminacy and judge bias. Within the LLM-as-a-judge literature (e.g., G-Eval Liu et al., 2023b, Prometheus Kim et al., 2024, and surveys like Gu et al., 2026), NL-PAC treats judge-mediated evaluation as a partially identified target-selection problem rather than a judge-alignment problem. Recent large-scale audits report that judge reliability and validity can come apart: high test–retest consistency can coexist with severe positional bias (Norman et al., 2026). Instrument-centric diagnostics likewise treat judge reliability as a property of the measurement instrument, not of individual outputs (Choi et al., 2026). This parallels NL-PAC’s treatment of the floor as a property of the tuple $(\text{LLM}_\theta, \Pi, \tau, P)$, not of any single judgment.

Guerdan et al. (2025) document *rating indeterminacy*, in which rating criteria admit multiple valid interpretations, and show that forced-choice judge validation is systematically biased relative to multi-label response sets; their contribution is a theoretical and empirical validation framework for judge systems, whereas NL-PAC proves and certifies the learnability floor that such indeterminacy imposes. Dorner et al. (2025) prove that, when the judge is no more accurate than the evaluated model, judge-based debiasing cannot reduce the required ground-truth sample size by more than half, and Feuer et al. (2026) study bias-bounded guarantees for measurable judge bias. Both quantify *exogenous* judge error; NL-PAC measures the endogenous ambiguity limit generated by the model’s own interpretation.

Learning from disagreement. The JAIR survey of Uma et al. (2021) organizes methods that retain multiple human judgments rather than reducing them to one gold label, and Baan et al. (2022) show why majority-label calibration is problematic when human disagreement is inherent. Perspectivist NLP develops the same contention into annotator- and distribution-aware modeling and evaluation (Xu and Jurgens, 2026). These approaches learn from or evaluate against human judgment variation. NL-PAC instead asks what can be guaranteed when the supervision channel does not reveal which retained reading is operative. Its validated reading family is therefore conceptually distinct from the model-admissible selector set: agreement between them is an empirical bridge condition, not a definition.

Prompt underspecification. Systems work on prompt sensitivity shows that prompts omit requirements that models fill with fragile defaults: underspecified prompts regress markedly across model or prompt changes (Yang et al., 2026), and even subjective evaluations can vary across prompt formulations. This continues the broader observation that underspecified pipelines admit many equally predictive solutions that diverge under deployment shift (D’Amour et al., 2022). Active task disambiguation (Kobalcyk et al., 2025) and elicitation (Li et al., 2025) clarify intent by asking questions or generating edge cases. NL-PAC supplies the complementary accounting layer by pricing the remaining ambiguity via D_τ^* .

Partial concept classes. Partial-concept PAC learning allows a concept to be undefined away from the support of its source distribution and studies which such classes remain learnable (Alon et al., 2022). NL-PAC has a different partiality: each admissible reading is a total target on the deployment domain, but the natural-language specification identifies a set of such targets and the mediated channel may fail to distinguish them. Thus the blind-channel floor is a channel-induced identification obstruction, not a consequence of undefined off-support labels.

Privileged information, semantic uncertainty, and partial-identification inference. Vapnik’s learning-using-privileged-information (LUPI) program treats teacher comments as training-only signals that accelerate learning (Vapnik and Vashist, 2009). While it asks how much trusted explanations help, our question is dual: how much error remains when the explanation channel is systematically ambiguous. The semantic-uncertainty literature (e.g., Kuhn et al., 2023) measures uncertainty over meanings by clustering answers; such concentration signals yield only one-sided diameter diagnostics, and the sharp handle for floors remains the master statistic D_τ^* . Finally, the certificate in Theorem B.3 draws on classical inference for partially identified parameters (Imbens and Manski, 2004) to certify the Le Cam floor from finite data; simultaneous-inference corrections over searched families in the sense of Berk et al. (2013) are deliberately unnecessary here, because the canonical candidate pair is a deterministic functional of the frozen model rather than a data-selected object.

Symbol	Defined	Meaning
$\mathcal{X}, \mathcal{Y}, P, f$	Section 2.1	instance space, finite label space, data distribution, target
f_{NL}	Section 2.1	natural-language task description (fixed string)
$\mathcal{T}, \mathcal{T}^*$	Section 2.1	token alphabet; string space under the discrete σ -algebra
$S_{\text{NL}}, J_{\text{NL}}$	Section 2.1	measurable string subspaces of $\mathcal{T}^*$
$\mathcal{F}_{S_{\text{NL}}}, \mathcal{F}_{J_{\text{NL}}}$	Section 2.1	their induced subspace σ -algebras
$h, \hat{h}$	Section 2.1	measurable classifier; learner (measurable rule)
g, ρ, j	Theorems 2.1 and 2.2	oracle kernel; justification decoder; justification
$\pi_{\text{LLM}_\theta}(y \mid f_{\text{NL}}, x)$	Section 2.1	admissibility score (description-only decoding probability)
τ, ζ	Theorem 2.8	admissibility threshold; coverage tolerance
$\mathcal{F}_{\tau, \zeta}(f_{\text{NL}})$	Theorem 2.8	model-admissible labeling class
$\mathcal{F}_{\tau, \zeta}^{\text{read}}(f_{\text{NL}})$	Theorem 3.6	coherent-reading subclass
$\text{diam}_{\tau, \zeta}(f_{\text{NL}}, P)$	Theorem 2.8	model-admissible diameter
d_P	Theorem 2.8	disagreement pseudometric $d_P(f, g) = \mathbb{P}_P[f \neq g]$
D, B	Theorem 3.2	pairwise disagreement mass; disagreement set
$A_\tau(x)$	Theorem 2.5	admissible label set at x
D_τ^*	Theorem 2.6	admissible-overlap mass $\mathbb{P}_x[A_\tau(x) \geq 2]$
V_τ^*	Theorem 3.4	exact blind-channel minimax on Sel_τ (Theorem 3.5); expected non-modal admissible mass
$V_{\text{blind}}(\cdot)$	Theorem 3.9	blind-channel minimax of a finite target family
$\mathcal{C}$	Theorem 3.9	finite coherent-reading family (distinct from $C = \mathcal{Y} $)
(f_τ^+, f_τ^-)	Theorem 2.9	maximal-spread pair
Sel_τ	Theorem 2.8	almost-everywhere admissible selection class; $\mathcal{F}_{\tau, \zeta}$ is its open ζ -neighborhood
$k(x)$	Section 3.2	number of admissible labels, $ \bar{A}_\tau(x) $
$\kappa(\xi)$	Theorem 4.5	threshold-margin mass
η	Theorem 3.8	coherent-family coverage gap
N	Theorem 4.1	held-out audit inputs (Hoeffding sample size)
m	Theorem 3.1	learner’s oracle-labeled sample size
r	Theorem 4.5	decoding depth (samples per input)
δ	Theorem 4.1	audit confidence level
ε_N	Theorem 4.1	Hoeffding sampling slack $\sqrt{\log(2/\delta)/2N}$
ξ	Theorem 4.5	threshold-margin half-width
C	Theorem 4.5	label-space size $ \mathcal{Y} $

Table 2: Recurring notation, with the defining location.

Task	Pair	D_{pop}	D_L	certified floor	$D_L/2$
Divisibility (control)	6 vs. 2 $\wedge$ 3	0.000	0.000		0.000
Pedagogy (ambiguous)	f_1 vs. f_2	0.400	0.332		0.166

Table 3: Controlled arithmetic and pipeline check of the finite-sample blind-channel floor certificate ($\tau = 0.20$, $\delta = 0.10$). D_{pop} is the known population disagreement fixed by construction, not an empirical estimate; D_L is the audited finite-sample lower bound.

Split	n	$\bar{H}$	$\mathbb{P}(k=3)$	$\frac{1}{2}D^*$	V^*	C_{half}	C_V
SNLI	1514	0.798	0.013	0.261	0.263	0.245	0.242
MNLI-matched	1599	1.072	0.059	0.376	0.386	0.361	0.366

Table 4: ChaosNLI multiclass check at $\tau = 0.20$ and $\delta = 0.10$, computed from the public ≈ 100 -annotation label distributions (no model). $C_{\text{half}} = \frac{1}{2}(\hat{D}^* - \varepsilon_N)_+$ and $C_V = (\hat{V}^* - \frac{2}{3}\varepsilon_N)_+$ are the two branches of Theorem 4.9. MNLI has enough three-way admissibility for the exact-value branch to improve the certified floor; SNLI does not. The comparison is conditional on the published human label distributions and is not a model calibration study.

Task	$\hat{D}^*$	$\hat{V}^*$	ε_N	κ_U	$2Ce^{-2r\xi^2}$	certified floor
Moderation (representative)	0.60	0.300	0.1358	1.0	1.0	0.0
Length control (matched)	0.05	0.025	0.1358	1.0	1.0	0.0

Table 5: Model-relative sampled-decoding audit of qwen2.5:3b ($N = 100$ audit and $N_\kappa = 100$ disjoint margin inputs, $r = 3$, $\tau = 0.2$, $\xi = 0.05$, $\delta = 0.1$), one matched prompt-seed configuration (paraphrase P1, seed 0); all twelve per-configuration cells are in Table 6. The certified floor is zero in every configuration because the finite-depth term is capped at 1 at $r = 3$.

Table 6: Per-configuration results for the model-relative Qwen 2.5-3B audit with $N = 100$, $N_\kappa = 100$, $r = 3$, $\tau = 0.2$, $\xi = 0.05$, and $\delta = 0.1$. “Majority frac.” denotes the pooled majority-label fraction; a value of 1.00 denotes constant output. Every certified floor is zero, as explained in the text, and the independently estimated bound κ_U enters the same radius.